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FSL

Fundamentals of statistical learning, rebuilt from scratch

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Common distributions

0.1 Normal distribution

Many phenomena in nature, like heights or measurement errors, pile up around a central value and become rarer and rarer as we move away from it, in the same way on both sides. The Normal, or Gaussian, distribution describes this bell shape, and it plays a central role in probability and statistics because many phenomena follow it, at least approximately.

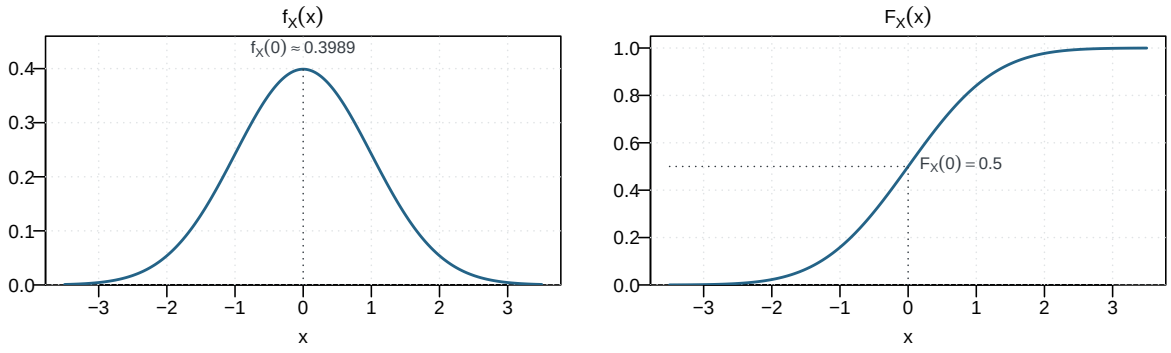
Definition 0.1: Normal distribution

A continuous random variable X has a **Normal** (or **Gaussian**) **distribution** with parameters $(\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$, written $X \sim N(\mu, \sigma^2)$, if its density and CDF are

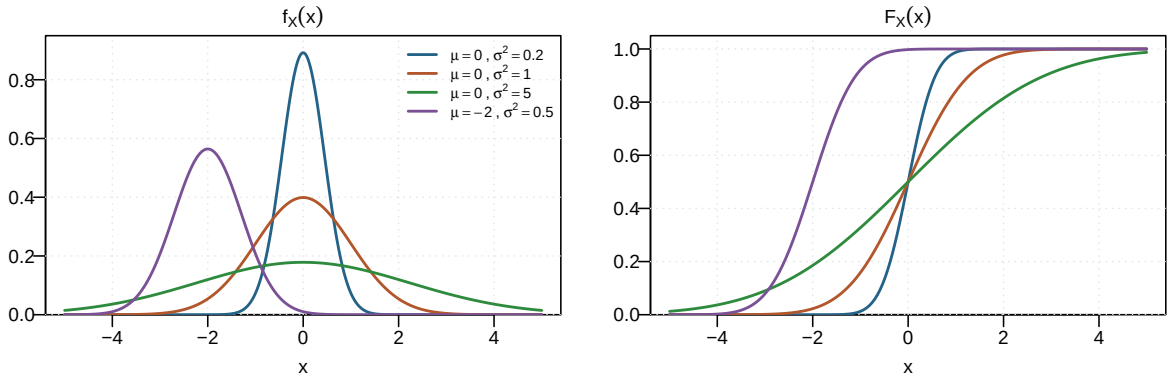
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t, \qquad x \in \mathbb{R}.$$

The CDF has no explicit form: the integral cannot be written with elementary functions, so we compute it numerically. The parameter μ is the *centre* of the distribution, and σ is its *spread*.

What does it look like? For $\mu = 0$ and $\sigma^2 = 1$ the density is a bell centred at 0, with its peak $f_X(0) = 1/\sqrt{2\pi} \approx 0.3989$. The CDF grows from 0 to 1 and, by symmetry, takes the value 0.5 at the centre.



Changing the parameters changes the bell in two different ways: μ moves it left or right, while σ^2 stretches it. A small σ^2 gives a tall, narrow bell and a steep CDF; a large σ^2 a low, wide bell and a CDF that grows slowly.



0.1.1 The standard Normal

Among all the Normals, one has a special name.

Definition 0.2: Standard Normal

X has a **standard Normal distribution** if $(\mu, \sigma^2) = (0, 1)$. A standard Normal random variable is traditionally denoted by Z, and its density and CDF by

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \qquad \Phi(z) = \int_{-\infty}^{z} \phi(t)\, \mathrm{d}t, \qquad z \in \mathbb{R}.$$

Why is it so important? Because every Normal can be turned into the standard one, and vice versa.

Proposition 0.1: Standardization

1. Any Normal can be turned into the standard Normal.

   If $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X - \mu}{\sigma} \sim N(0, 1)$.

2. The standard Normal can be turned into any Normal.

Conversely, if $Z \sim N(0,1)$, then $X = \mu + \sigma Z \sim N(\mu, \sigma^2)$.

3. The quantile of a Normal is obtained by taking the one of the standard Normal and transforming it like any other value, that is, multiplying by σ and adding μ.

   If q_p and z_p are the quantiles of level p of $N(\mu, \sigma^2)$ and of $N(0,1)$, then $q_p = \mu + z_p\,\sigma$.

Proof (point 1). To show that $Z \sim N(0,1)$, we go in three steps:

1. we start from the CDF of Z;
2. we differentiate it to get the density f_Z;
3. we compare f_Z with ϕ, which has an explicit formula.

If the densities coincide, so do the CDFs, since the CDF is the integral of the density.

Since $\sigma > 0$, multiplying by σ and adding μ on both sides keeps the inequality, so

$$F_Z(z) = P(Z \le z) = P\Big(\frac{X-\mu}{\sigma} \le z\Big) = P(X \le \mu + \sigma z) = F_X(\mu + \sigma z).$$

Differentiating with respect to z gives

$$f_Z(z) = f_X(\mu + \sigma z)\,\sigma.$$

Finally we evaluate the Normal density at $x = \mu + \sigma z$:

$$f_Z(z) = \sigma \cdot \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(\mu+\sigma z-\mu)^2}{2\sigma^2}} = \sigma \cdot \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{\sigma^2 z^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{z^2}{2}} = \phi(z),$$

which is the density of $N(0,1)$. □

Proof (point 2). To show that $X \sim N(\mu, \sigma^2)$, we go in three steps:

1. we start from the CDF of X;
2. we differentiate it to get the density f_X;
3. we compare f_X with the density of the Normal distribution, which has an explicit formula.

If the densities coincide, so do the CDFs, since the CDF is the integral of the density.

Since $\sigma > 0$, subtracting μ and dividing by σ on both sides keeps the inequality, so

$$F_X(x) = P(X \le x) = P(\mu + \sigma Z \le x) = P\Big(Z \le \frac{x-\mu}{\sigma}\Big) = \Phi\Big(\frac{x-\mu}{\sigma}\Big).$$

Differentiating with respect to x gives

$$f_X(x) = \phi\Big(\frac{x-\mu}{\sigma}\Big)\,\frac{1}{\sigma}.$$

Finally we evaluate the standard Normal density at $z = (x-\mu)/\sigma$:

$$f_X(x) = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

which is the density of $N(\mu, \sigma^2)$. □

Proof (point 3). By point 2, every value z of Z becomes the value $\mu + \sigma z$ of X. Point 3 says that the quantile z_p becomes the quantile q_p. Since q_p is, by definition, the point that leaves area p on its left under $N(\mu, \sigma^2)$, we check that the candidate $\mu + \sigma z_p$ has this property:

1. we write $X = \mu + \sigma Z$, by point 2;
2. since $\sigma > 0$, subtracting μ and dividing by σ on both sides keeps the inequality;
3. z_p leaves area p on its left, that is, $\Phi(z_p) = p$.

Then

$$P(X \le \mu + \sigma z_p) \overset{1}{=} P(\mu + \sigma Z \le \mu + \sigma z_p) \overset{2}{=} P(Z \le z_p) \overset{3}{=} p,$$

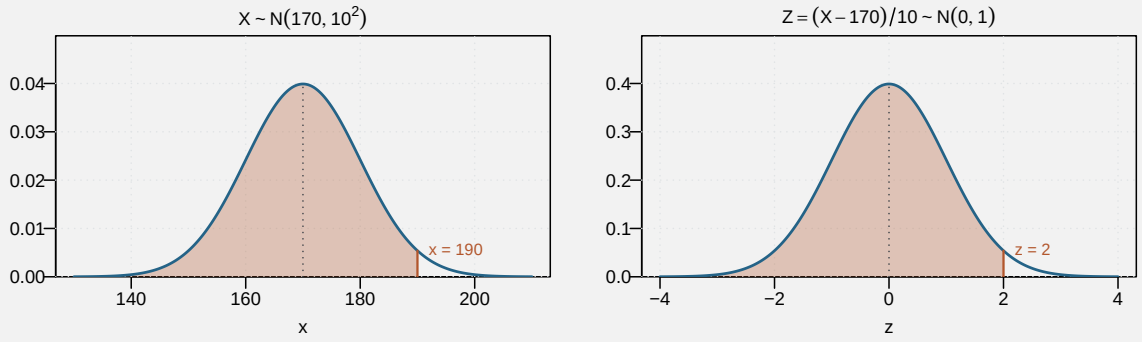
so the candidate is the quantile: $q_p = \mu + z_p\,\sigma$. □

Example 1: Standardizing the heights

Let us see an example of the first point of Proposition 0.1.

Let the heights be $X \sim N(170, 10^2)$. Standardizing moves the bell from 170 to 0 and shrinks its width from 10 to 1.

A student 190 cm tall becomes $z = (190 - 170)/10 = 2$: that is, the student lies $2\sigma = 20$ cm above the centre μ.



To study later

0.1.2 Normal probabilities

We want to compute $P(a < X < b)$ for any Normal $X \sim N(\mu, \sigma^2)$. But the CDF of the Normal has no explicit form, so there is no formula to compute by hand. For the standard Normal, however, the values of Φ are tabulated.

The idea is to use point 1 of Proposition 0.1: we bring the probability back to Z, and then use Φ. We go in three steps:

1. we standardize all three sides, subtracting μ and dividing by σ; since $\sigma > 0$, the event does not change;
2. in the middle there is Z, since $\frac{X-\mu}{\sigma} = Z \sim N(0,1)$ by point 1;
3. the area between two points is the CDF at the right point minus the CDF at the left point.

Then

$$P(a < X < b) \overset{1}{=} P\Big(\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\Big) \overset{2}{=} P\Big(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\Big)$$
$$\overset{3}{=} \Phi\Big(\frac{b-\mu}{\sigma}\Big) - \Phi\Big(\frac{a-\mu}{\sigma}\Big).$$

For any Normal it is enough to know Φ: we standardize the endpoints a and b, and read Φ at those two points.

Example 2: Normal probabilities

Let $X \sim N(3,5)$. What is $P(X > 1)$? Standardizing,

$$P(X > 1) = 1 - P(X < 1) = 1 - P\Big(Z < \frac{1-3}{\sqrt{5}}\Big) = 1 - \Phi(-0.8944)$$
$$= 1 - \texttt{pnorm(-0.8944)} = 0.8144.$$

0.1.3 Normal quantiles

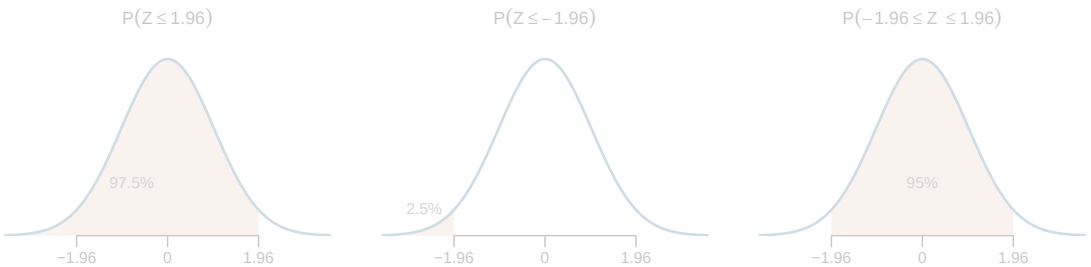
The other way round, we often need the value below which a given probability lies. Let $Z \sim N(0,1)$ and let z_p be its quantile of level p, the value that solves $p = \Phi(z_p)$. Some quantiles are used so often, in confidence intervals and statistical tests, that they are worth knowing:

p	0.9000	0.9500	0.9750	0.9900	0.9950	0.9975
z_p	1.28	1.64	1.96	2.33	2.58	2.81

For example, $z_{0.975} \approx 1.96$. Since the density ϕ is symmetric around 0,

$$z_p = -z_{1-p},$$

so -1.96 leaves 2.5% of the area on its left, and 95% of the area under ϕ lies between -1.96 and 1.96.



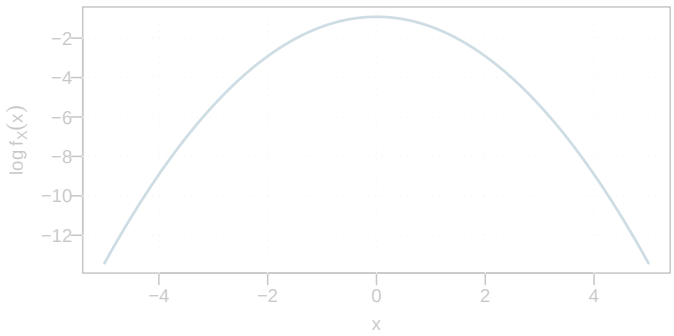
More generally, the shortest interval that contains probability $1-\alpha$ is delimited by the quantiles $z_{\alpha/2}$ and $z_{1-\alpha/2}$.

0.1.4 Heads-up: light tails

The Normal is often presented as the most common distribution in the real world, but it is a special, rather extreme, unimodal distribution. How fast does its density go to 0? Up to the constant,

$$f_X(x) \propto e^{-\frac{1}{2}x^2} \qquad \text{for } X \sim N(0,1),$$

so the log-density $-\frac{1}{2}x^2$ is a parabola: it only decays quadratically with the distance from the centre. The density itself goes to 0 as $e^{-x^2/2}$ when $|x| \to +\infty$, extremely fast, so the probability of seeing something in the tails is tiny. The Normal has **light tails**.



How tiny? The typical deviation of X from the centre μ is σ, but the probabilities that X deviates from μ by more than 2σ, 3σ, 4σ and 5σ are

k	2	3	4	5
$P(\lvert X-\mu\rvert > k\sigma)$	0.046	0.003	$6\cdot10^{-5}$	$6\cdot10^{-7}$

Deviations four or five times larger than the typical one may be rare in a real population, but not as rare as $6\cdot10^{-5}$. If we model with a Normal a variable that actually has heavier tails, the rest of the model will force itself to reduce the deviations of the outliers. The Normal is too sensitive to outliers, and we need distributions with heavier tails.

0.2 Student's t distribution

Definition 0.3: Student's t distribution

X has a **Student's t distribution** with $\nu > 0$ degrees of freedom, written $X \sim t_\nu$, if its density is

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \qquad x \in \mathbb{R}.$$

0.2.1 Cauchy distribution

With $\nu = 1$ the constant becomes $\Gamma(1)/\left(\sqrt{\pi}\,\Gamma(\frac{1}{2})\right) = 1/\pi$, and the density takes a particularly simple form.

Definition 0.4: Cauchy distribution

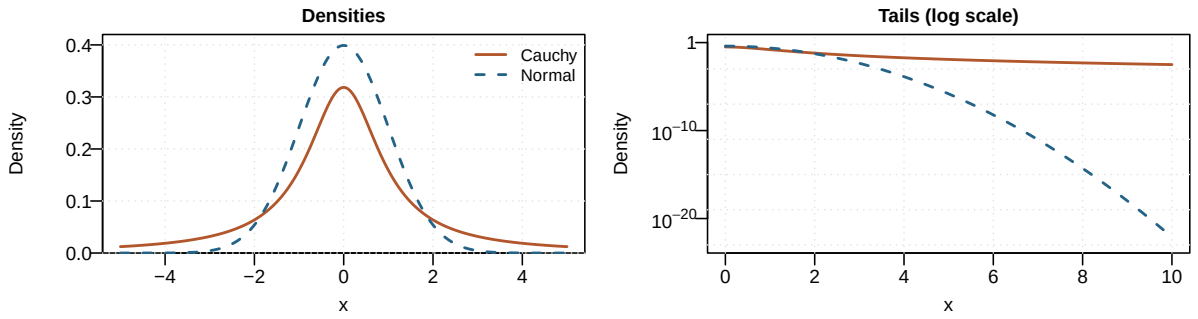
X has a **Cauchy distribution** if its density is

$$f_X(x) = \frac{1}{\pi\,(x^2+1)}, \qquad x \in \mathbb{R},$$

that is, if $X \sim t_1$.

0.2.1.1 A first look

The density is symmetric about 0, because replacing x with $-x$ leaves x^2 unchanged, and it peaks at $f_X(0) = 1/\pi \approx 0.32$. Its bell shape resembles the one of the standard Normal (Definition 0.1), but the tails are very different. For large x, $f_X(x)$ behaves like $1/(\pi x^2)$, which decreases only as a power of x, while the standard Normal density decreases like $e^{-x^2/2}$, way faster. At $x = 10$ the Cauchy density is about 0.003, the standard Normal one about $8 \cdot 10^{-23}$. Values far from 0 are unlikely under the Cauchy, but not negligible: it has **heavy tails**.



Same bell shape near 0; on a log scale the Normal tail collapses while the Cauchy tail stays high.

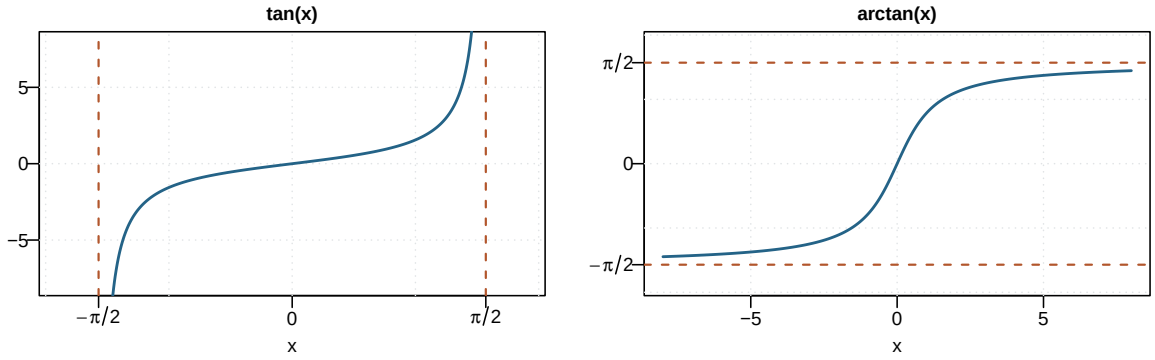
0.2.1.2 Check if $\int_{\mathbb{R}} f_X(x)\,\mathrm{d}x = 1$

We need a primitive of $1/(x^2+1)$, and calculus gives one:

$$\frac{\mathrm{d}}{\mathrm{d}x}\arctan(x) = \frac{1}{x^2+1}, \qquad x \in \mathbb{R}.$$

The arctangent is the inverse of the tangent restricted to $(-\pi/2, \pi/2)$: the tangent maps that interval onto the whole real line, so the arctangent maps $\mathbb{R}$ back onto $(-\pi/2, \pi/2)$, with

$$\lim_{x\to-\infty}\arctan(x) = -\frac{\pi}{2}, \qquad \lim_{x\to+\infty}\arctan(x) = \frac{\pi}{2}.$$



Hence

$$\int_{\mathbb{R}} \frac{1}{\pi\,(x^2+1)}\,\mathrm{d}x = \frac{1}{\pi}\Big[\arctan(x)\Big]_{-\infty}^{+\infty} = \frac{1}{\pi}\Big(\frac{\pi}{2} - \Big(-\frac{\pi}{2}\Big)\Big) = 1.$$

1 Learning

1.1 The learning problem

Why do we collect data? Because we want to understand the process that produced them. Suppose we want to study the heights of the students of a university. The heights of all the students have some distribution, with many values around 170 cm and few below 150 or above 200. This distribution, with CDF F_X, is the **population model**: it describes what varies in the population and how. In a learning problem, however, we do not know it.

We cannot measure everyone, so we pick n students. Before measuring them, their heights are random variables $X_1, \ldots, X_n$, where X_i is the height of the i-th student we pick, since we do not know yet who will be picked. The **sampling scheme** describes how we pick them, and the simplest one is IID:

- **independent**: the height of the first student tells us nothing about the height of the second;
- **identically distributed**: each X_i has the population distribution F_X, since each student comes from the same population.

The population model and the sampling scheme together form the **data-generating process**, the machine that produces the numbers we see. Once we measure, we have the observed data

$$\mathcal{D}_n = (x_1, \ldots, x_n).$$

For example, with $n = 50$,

$$\mathcal{D}_{50} = (188,\ 164,\ 167,\ 169,\ 165,\ 165,\ 177,\ 171,\ \ldots,\ 169,\ 181,\ 176,\ 173,\ 179).$$

The uppercase X_i is the random variable before measuring; the lowercase x_i is the number after. In short,

$$\underbrace{F_X}_{\text{population (unknown)}} \xrightarrow{\text{IID sampling}} \underbrace{X_1, \ldots, X_n}_{\text{random}} \longrightarrow \underbrace{x_1, \ldots, x_n}_{\text{observed data}}.$$

Probability goes from left to right: if we know F_X, we can say which data will come out.

Learning goes in the other direction: we start from the observed data and try to recover the population distribution:

$$\underbrace{F_X}_{?} \longleftarrow \underbrace{x_1, \ldots, x_n}_{\text{known}}.$$

In practice we choose a **model**, a family of distributions p_θ described by a **parameter** θ, which may be one number or several. Choosing a value of θ chooses one candidate distribution, and learning means choosing the value that explains the data best.

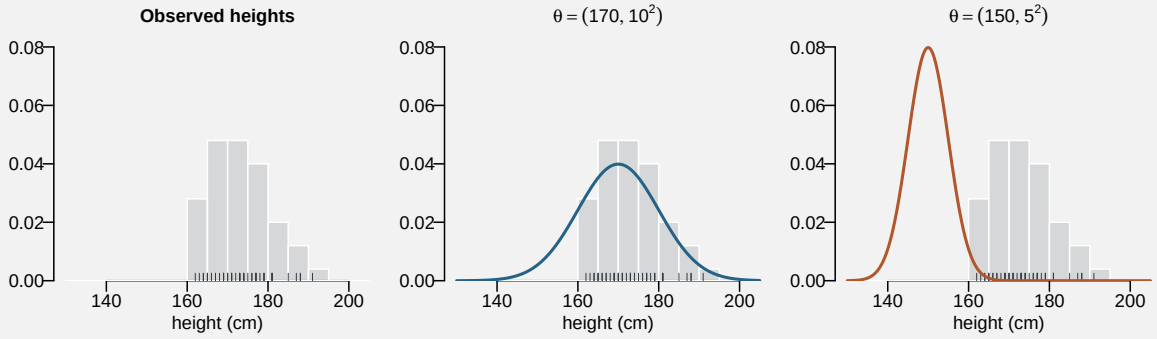
Example 3: Heights as a Normal model

For the heights we may assume a Normal distribution, without knowing which one. Recall from Definition 0.1 that a Normal is fixed by its centre μ and its spread σ, so

$$\theta = (\mu, \sigma^2).$$

Let us take the $n = 50$ heights $\mathcal{D}_{50}$ we measured above. The first plot shows them as a histogram, with a small mark below for each student: they lie between 162 and 191 cm, mostly around 170–175.

Each value of θ is a different candidate: $\theta = (170, 10^2)$ is a Normal centred at 170, while $\theta = (150, 5^2)$ is centred at 150 and narrower. How well does each one approximate the data? We draw it over the histogram.



The first candidate follows the histogram closely. The second puts almost all its probability below 165 cm, where we observed almost no students, and almost none where most of them are. The first candidate explains the data much better than the second.

How do we compare the candidates? The likelihood is exactly what we need.

Definition 1.1: Likelihood and maximum likelihood

The **likelihood** of θ is the joint density of the observed data, seen as a function of θ. For IID data it is the product

$$L(\theta) = \prod_{i=1}^{n} p_\theta(x_i).$$

The **maximum likelihood estimate** is the value that makes it largest:

$$\hat{\theta} = \arg\max_\theta \; L(\theta).$$

The likelihood is a product of n factors. Taking the logarithm turns it into a sum, since $\log(ab) = \log a + \log b$.

Definition 1.2: Log-likelihood

The **log-likelihood** is the logarithm of the likelihood. For IID data it is the sum

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p_\theta(x_i).$$

Why is it convenient?

1. **Sums are easy to differentiate, products are not.** To find the maximum we differentiate and set the derivative to 0. The derivative of a sum is the sum of the derivatives, while the derivative of a product is more complex.
2. **No tiny numbers.** The likelihood multiplies n values that are often smaller than 1, so it shrinks quickly as n grows and soon becomes too small to handle, even for a computer. The log-likelihood adds their logarithms instead, so it stays an ordinary number.

Why do the likelihood and the log-likelihood have the same maximum? Let $\hat{\theta}$ maximise L, so that

$$L(\hat{\theta}) \geq L(\theta) \qquad \text{for every } \theta.$$

Since log is increasing, applying it to both sides keeps the inequality:

$$\log L(\hat{\theta}) \geq \log L(\theta),$$

that is,

$$\ell(\hat{\theta}) \geq \ell(\theta) \qquad \text{for every } \theta,$$

so $\hat{\theta}$ maximises ℓ too. The same argument works backwards, with the exponential, which is increasing as well. Hence

$$\hat{\theta} = \arg\max_\theta L(\theta) = \arg\max_\theta \ell(\theta) :$$

the values change, the point of maximum does not.

For a Normal model the maximum can be found by hand.

Proposition 1.1: Maximum likelihood for a Normal

Let $x_1, \dots, x_n$ be IID data from $N(\mu, \sigma^2)$. The maximum likelihood estimates of μ and σ^2 are

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Proof. The density of a single observation is

$$p_\theta(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}.$$

Its logarithm splits the product into a sum, and undoes the exponential, since $\log e^a = a$:

$$\log p_\theta(x_i) = \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{(x_i-\mu)^2}{2\sigma^2} = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x_i-\mu)^2}{2\sigma^2},$$

where we used $\log \frac{1}{\sqrt{a}} = -\frac{1}{2}\log a$. The log-likelihood is the sum over the n observations; the first term is the same for all of them, so it appears n times:

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

We set the derivatives to 0. The first term does not contain μ, so

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x},$$

since the equation is linear in μ. Then

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 = 0 \quad \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2.$$

□

The best centre is the mean of the data, and the best spread measures how far the data are from it.

Example 3 part 2: Heights as a Normal model

Let us take the heights of Example 3 again. From the plots, the first candidate explains the data much better than the second. Does the likelihood say the same?

First we evaluate each density at every observed value:

x_i	$\theta = (170, 10^2)$	$\theta = (150, 5^2)$
188	0.0079	$2 \cdot 10^{-14}$
164	0.033	0.0016
167	0.038	0.00025
⋮	⋮	⋮

Then we multiply the 50 values:

$$L(170, 10^2) \approx 2 \cdot 10^{-77}, \qquad L(150, 5^2) \approx 9 \cdot 10^{-319}.$$

Both numbers are tiny, since we multiply 50 values smaller than 1; what matters is the comparison:

$$\frac{L(170, 10^2)}{L(150, 5^2)} \approx 10^{241}.$$

The likelihood agrees with the plots.

Maximum likelihood looks for the largest L among all the candidates, not only these two. By Proposition 1.1, for these data it is reached at $\hat{\theta} = (\bar{x}, \hat{\sigma}^2) \approx (173.6,\ 7.0^2)$, with $L(\hat{\theta}) \approx 9 \cdot 10^{-74}$.

Summing up, learning goes in four steps.

Algorithm 1.1: Learning a model from data

1. **Data.** Collect IID data $\mathcal{D}_n = (x_1, \ldots, x_n)$.
2. **Model.** Choose a family of distributions p_θ, with parameter θ.
3. **Likelihood.** Write the likelihood of the data, $L(\theta) = \prod_{i=1}^{n} p_\theta(x_i)$.
4. **Maximum likelihood.** Choose $\hat{\theta} = \arg\max_\theta L(\theta)$, or equivalently $\arg\max_\theta \ell(\theta)$.

Example 3 and its part 2 follow exactly these steps:

1. **Data:** the $n = 50$ heights $\mathcal{D}_{50}$;
2. **Model:** a Normal, with $\theta = (\mu, \sigma^2)$;
3. **Likelihood:** the product of the Normal densities at the 50 heights;
4. **Maximum likelihood:** $\hat{\theta} \approx (173.6,\ 7.0^2)$.

1.2 Hidden variables and mixtures

In Section 1.1 each observation x_i was a height, and the model p_θ described the heights directly. Often, however, behind what we observe, like an image, a symbol or an audio recording, there is something we do not see, which affects what we see: a **hidden** variable.

For example, we measure the height of a person without knowing whether the person is a man or a woman. The sex matters, since men are taller on average, but we do not observe it. Then

- obs is the height, what we see;
- hid is the sex, what is hidden.

We describe the two together with a joint distribution $p(\text{obs}, \text{hid})$. By the product rule it splits into two pieces:

$$\begin{bmatrix}\text{obs}\\ \text{hid}\end{bmatrix} \sim \underbrace{p(\text{obs}, \text{hid})}_{\text{joint}} = \underbrace{p(\text{obs} \mid \text{hid})}_{\text{conditional}}\, \underbrace{p(\text{hid})}_{\text{marginal}}.$$

We can read it as a story in two steps: first the hidden value is drawn with probability $p(\text{hid})$, then, given that value, the observation is drawn from $p(\text{obs} \mid \text{hid})$.

What is the distribution of what we actually see? We do not know the hidden value, so we integrate it out over its support $\text{supp}(\text{hid})$, the set of all the values it can take:

$$p(\text{obs}) = \int_{\text{supp}(\text{hid})} p(\text{obs}, \text{hid})\, \mathrm{d}\,\text{hid}.$$

Example 4: Heights of men and women

In the height example the support of the hidden variable is $\text{supp}(\text{hid}) = \{\text{man}, \text{woman}\}$, only two values, so the integral is a sum over the two groups:

$$p(\text{height}) = p(\text{height} \mid \text{man})\, p(\text{man}) + p(\text{height} \mid \text{woman})\, p(\text{woman}).$$

So $p(\text{obs})$ is a weighted average of several distributions, one for each hidden value, weighted by how likely that value is: it is a **mixture**.

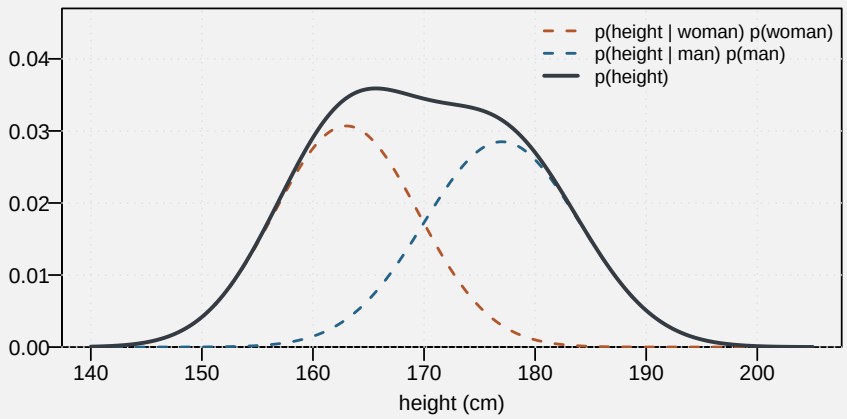
Example 4 part 2: Heights of men and women

Let us take the heights of Example 4 again, where $p(\text{height})$ is a sum over the two groups. Why is it a weighted average of distributions? Let us label each piece:

$$p(\text{height}) = \underbrace{p(\text{height} \mid \text{man})}_{\text{distribution of the men}} \cdot \underbrace{p(\text{man})}_{\text{weight}} + \underbrace{p(\text{height} \mid \text{woman})}_{\text{distribution of the women}} \cdot \underbrace{p(\text{woman})}_{\text{weight}}.$$

For each hidden value there is a distribution of the heights: a bell for the men and one for the women. Each bell is multiplied by the probability of its group, and the weights add up to $p(\text{man}) + p(\text{woman}) = 1$. So $p(\text{height})$ is a weighted average of the two bells.

In the plot below the two groups have the same weight, $p(\text{man}) = p(\text{woman}) = 0.5$.



1.2.1 Finite mixtures

In Example 4 the hidden variable had only two values, man and woman. What happens if the population is made of K groups instead of two?

We make two assumptions:

1. The hidden variable hid has a finite support, $\text{supp}(\text{hid}) = \{1, \dots, K\}$: each value is a group, or **sub-population**. The probability of each group is, again, a **weight**,

$$\pi_k = P(\text{hid} = k), \qquad k = 1, \dots, K.$$

   The weights are probabilities, so they are non-negative and add up to 1. Vectors like this have a name.

Definition 1.3: Simplex

The **simplex** of dimension $K - 1$ is the set

$$\Delta^{K-1} = \left\{ (p_1, \dots, p_K) \in \mathbb{R}^K \;\middle|\; p_i \geq 0,\ \sum_{i=1}^{K} p_i = 1 \right\}.$$

   So the vector of weights lies in the simplex: $(\pi_1, \dots, \pi_K) \in \Delta^{K-1}$.

2. Inside each group the observation has its own distribution, usually from a simple family:

$$(\text{obs} \mid \text{hid} = k) \sim f_k(o).$$

   It may be discrete or continuous, univariate or multivariate.

We are still interested in computing the distribution of what we actually see. As in Section 1.2, we sum out the hidden variable:

$$f_{\text{obs}}(o) = \int_{\text{supp(hid)}} p(o,\, h)\, \mathrm{d}h.$$

Since hid takes only the values $1, \dots, K$, the integral becomes a sum:

$$f_{\text{obs}}(o) = \sum_{k=1}^{K} p(o, \text{hid} = k).$$

Each term splits, by the multiplication rule, into a conditional and a marginal:

$$p(o, \text{hid} = k) = \underbrace{p(o \mid \text{hid} = k)}_{f_k(o) \text{ (assumption 2)}} \cdot \underbrace{P(\text{hid} = k)}_{\pi_k \text{ (assumption 1)}}.$$

Since we know both pieces on the right-hand side from the two assumptions, putting them together gives the following model.

Definition 1.4: Finite mixture model

A **finite mixture model** with K **components** has density (or mass function)

$$f_{\text{obs}}(o) = \sum_{j=1}^{K} f_j(o)\, \pi_j,$$

where $f_1, \dots, f_K$ are the distributions of the K groups and $(\pi_1, \dots, \pi_K) \in \Delta^{K-1}$ are their weights.

Example 4 part 3: Heights of men and women

Let us take the heights of Example 4 again, where

$$p(\text{height}) = p(\text{height} \mid \text{man})\, p(\text{man}) + p(\text{height} \mid \text{woman})\, p(\text{woman}).$$

Is it a finite mixture? Yes, with $K = 2$ groups: the hidden variable hid takes the values woman and man, with weights and distributions

$$\pi_1 = p(\text{woman}), \quad \pi_2 = p(\text{man}), \qquad f_1 = p(\text{height} \mid \text{woman}), \quad f_2 = p(\text{height} \mid \text{man}).$$

1.2.1.1 Gaussian mixtures

The finite mixture leaves the distribution f_k of each group free: it can be anything. The most common choice is a Normal in every group, each with its own centre μ_j and its own spread σ_j:

$$f_j(o) = N(o \mid \mu_j, \sigma_j^2), \qquad j = 1, \dots, K.$$

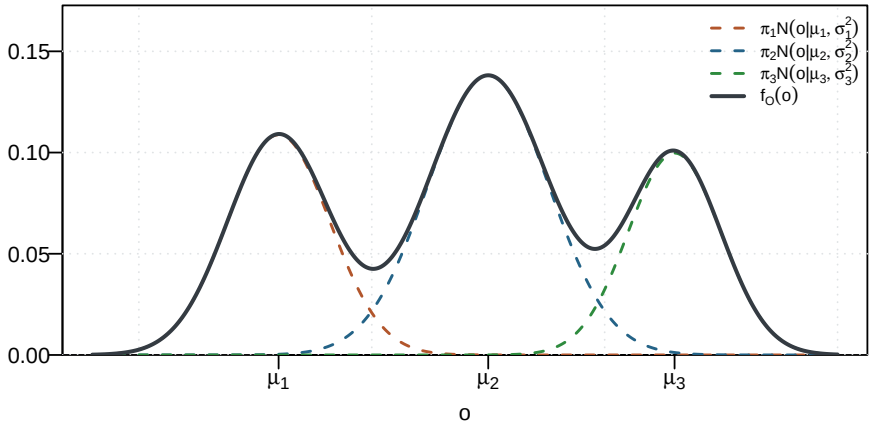
Definition 1.5: Gaussian mixture model

A **Gaussian mixture model** (GMM) with K components has density

$$f_{\text{obs}}(o) = \sum_{j=1}^{K} \pi_j\, N(o \mid \mu_j, \sigma_j^2),$$

with parameters $\theta = \{\pi_j,\, \mu_j,\, \sigma_j^2\}_{j=1}^K$, where $(\pi_1, \ldots, \pi_K) \in \Delta^{K-1}$.

A single Normal always has one bump and is symmetric. By adding K weighted bells we can describe data with several bumps, or asymmetric ones. For example, with $K = 3$:



Example 4 part 4: Heights of men and women

Let us take the heights of Example 4 part 3 again: a finite mixture with $K = 2$ groups, women and men, with weights π_1, π_2 and distributions $f_1 = p(\text{height} \mid \text{woman})$, $f_2 = p(\text{height} \mid \text{man})$. Is it a Gaussian mixture?

It is, if the heights inside each group are Normal:

$$f_1 = N(\mu_1, \sigma_1^2), \qquad f_2 = N(\mu_2, \sigma_2^2).$$

Then

$$p(\text{height}) = \pi_1\, N(\text{height} \mid \mu_1, \sigma_1^2) + \pi_2\, N(\text{height} \mid \mu_2, \sigma_2^2),$$

a GMM with $K = 2$ and parameters $\theta = \{\pi_1, \pi_2, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2\}$.

How do we learn the θ of a GMM from data? Let us follow the steps of Algorithm 1.1, writing the data as $o_1, \ldots, o_n$, the observed values of obs:

1. **Data:** IID observations $o_1, \ldots, o_n$;
2. **Model:** the GMM of Definition 1.5, with $\theta = \{\pi_j, \mu_j, \sigma_j^2\}_{j=1}^K$;
3. **Likelihood:** as always $L(\theta) = \prod_{i=1}^n p_\theta(o_i)$, but now each factor is the density of the GMM,

$$p_\theta(o_i) = \sum_{j=1}^K \pi_j\, N(o_i \mid \mu_j, \sigma_j^2),$$

so that

$$L(\theta) = \prod_{i=1}^n \sum_{j=1}^K \pi_j\, N(o_i \mid \mu_j, \sigma_j^2);$$

4. **Maximum likelihood:** $\hat{\theta} = \arg\max_\theta L(\theta)$.

Steps 1, 2 and 3 work as before. Step 4 is where we get stuck. Why?

To see why, consider first the case where we know the group of every observation: its hidden label hid_i takes an observed value h_i.

Proposition 1.2: Maximum likelihood for a GMM with labels

Let $o_1, \dots, o_n$ be IID data from a GMM with K components, and suppose that the group h_i of every observation is known. Let n_j be the number of observations in group j. Then the maximum likelihood estimates are

$$\hat{\pi}_j = \frac{n_j}{n}, \qquad \hat{\mu}_j = \frac{1}{n_j} \sum_{i:\, h_i = j} o_i, \qquad \hat{\sigma}_j^2 = \frac{1}{n_j} \sum_{i:\, h_i = j} (o_i - \hat{\mu}_j)^2, \qquad j = 1, \dots, K.$$

Proof. Recall that, without labels, the likelihood of a GMM is

$$L(\theta) = \prod_{i=1}^{n} \sum_{j=1}^{K} \pi_j \, N(o_i \mid \mu_j, \sigma_j^2).$$

With the labels we know that it is h_i, so only the term $j = h_i$ of the sum remains:

$$L(\theta) = \prod_{i=1}^{n} \pi_{h_i} \, N(o_i \mid \mu_{h_i}, \sigma_{h_i}^2).$$

Taking the logarithm, the product over i becomes a sum, and each factor splits since $\log(ab) = \log a + \log b$:

$$\ell(\theta) = \sum_{i=1}^{n} \Big[\log \pi_{h_i} + \log N(o_i \mid \mu_{h_i}, \sigma_{h_i}^2) \Big].$$

Grouping the terms by group, it breaks into separate pieces:

$$\ell(\theta) = \underbrace{\sum_{j=1}^{K} n_j \log \pi_j}_{\text{only the weights}} + \sum_{j=1}^{K} \underbrace{\sum_{i:\, h_i = j} \log N(o_i \mid \mu_j, \sigma_j^2)}_{\text{only } \mu_j \text{ and } \sigma_j^2}.$$

The parameters μ_j and σ_j^2 appear only in the j-th inner sum, which is the log-likelihood of a single Normal on the n_j observations of group j. By Proposition 1.1, its maximum is at the mean and the spread of the group. The weights appear only in the first sum; maximising it under the constraint $\sum_j \pi_j = 1$ gives, as one can show, $\hat{\pi}_j = n_j/n$. □

Example 4 part 5: Heights of men and women

Let us take the heights of Example 4 part 4 again: a GMM with $K = 2$ groups, women and men. Can we learn its θ following Algorithm 1.1?

With labels

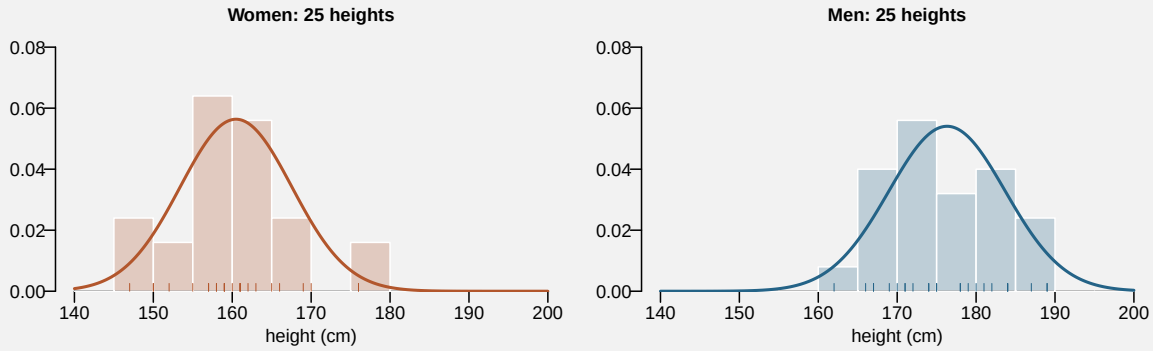
Suppose first that we measured 50 people and recorded, next to each height, also the group:

height	group
147	woman
166	woman
182	man
184	man
167	man
⋮	⋮

Then learning would be easy: we are in the case of Proposition 1.2, with $K = 2$, $n_1 = 25$ women and $n_2 = 25$ men. In practice we split the data in two and on each part we learn a single Normal, exactly as we did for all the heights in Example 3 and its part 2. For these data,

$$\hat{\theta}_{\text{women}} \approx (160.5,\, 7.1^2), \qquad \hat{\theta}_{\text{men}} \approx (176.3,\, 7.4^2),$$

and each fitted Normal follows its own histogram:



The weights are the fractions of each group:

$$\hat{\pi}_1 = \frac{n_1}{n} = \frac{25}{50} = 0.5, \qquad \hat{\pi}_2 = \frac{n_2}{n} = \frac{25}{50} = 0.5.$$

Without labels

In practice, however, the column "group" is missing: we only see the heights. So every factor of the likelihood is the sum of the two bells:

$$L(\theta) = \prod_{i=1}^{n} \Big[\pi_1\, N(o_i \mid \mu_1, \sigma_1^2) + \pi_2\, N(o_i \mid \mu_2, \sigma_2^2)\Big].$$

The log-likelihood of Definition 1.2 still turns the product into a sum, but inside each term a logarithm of a sum remains:

$$\ell(\theta) = \sum_{i=1}^{n} \log \Big(\pi_1 \, N(o_i \mid \mu_1, \sigma_1^2) + \pi_2 \, N(o_i \mid \mu_2, \sigma_2^2) \Big).$$

This is where we get stuck: $\log(a+b) \neq \log a + \log b$, so the parameters of the women and of the men stay tangled together in every term, and they no longer separate.

What happens if we set a derivative to 0 anyway? For μ_1,

$$\frac{\partial \ell}{\partial \mu_1} = \sum_{i=1}^{n} \frac{\pi_1 N(o_i \mid \mu_1, \sigma_1^2)}{\pi_1 N(o_i \mid \mu_1, \sigma_1^2) + \pi_2 N(o_i \mid \mu_2, \sigma_2^2)} \, \frac{o_i - \mu_1}{\sigma_1^2} = 0.$$

Here μ_1 appears both in $o_i - \mu_1$ and inside the exponentials of the Normal densities, in the numerator and in the denominator. Unlike the linear equation of Proposition 1.1, no algebraic step isolates μ_1: there is no formula for $\hat{\theta}$. The maximum still exists, but we can only approach it step by step.

To fit each group we would need to know which observations belong to it, and to know that we would already need the parameters.

Important. With the tools we have so far, we cannot compute the maximum likelihood estimate of a GMM without labels. Later, once we know more powerful tools, we will be able to solve this problem, which we call (♠).

2

Expected value

2.1 Why summarise a distribution?

The distribution of a random variable X tells us which values X can take and how likely each of them is. It contains *all* the probabilistic information about X: once we know it, we can answer any question such as "what is the probability that a light bulb lasts more than 1000 hours?" or "that it lasts between 500 and 800 hours?".

That completeness has a price. As we already know, the distribution is a *function*: the probability mass function $p_X(x)$ if X is discrete, the density $f_X(x)$ if it is continuous, or the cumulative distribution function $F_X(x)$ in either case. It assigns a value to every possible x, often to infinitely many of them. If we ask "roughly how long does this bulb last?" and receive a whole density curve, we have all the information but no answer. Can we describe X with a few numbers instead of a whole function, accepting that we lose some information?

2.1.1 Median and quantiles

The simplest *summaries* pick a few points of the distribution and give up describing the entire curve. We have already met them with the quantile function, so we only recall them quickly.

The **median** is the value that splits the distribution in half: X falls below it with probability $1/2$ and above it with probability $1/2$. If the median lifetime of a bulb is 900 hours, half of the bulbs fail before 900 hours and half after.

The **quantile** of order p extends the idea to any proportion: it is the value below which X falls with probability p, that is the point where F_X reaches p. The quantile of order 0.5 is the median, while the quantiles of order 0.25 and 0.75 are the first and third quartiles. If the quartiles of the bulb lifetime are 600 and 1200 hours, the central half of the bulbs lasts between 600 and 1200 hours, so three numbers already describe both the centre and the spread.

They remain summaries, though: two very different distributions, one with very long tails and one without, can share the same median and quartiles.

2.1.2 The expected value as a weighted average

There is another way to condense X into a single number, which uses *all* its values at once.

Definition 2.1: Expected value: the intuitive idea

The **expectation**, **expected value** or **mean** $\mathbb{E}(X)$ is the weighted average of the possible values of X, with weights equal to their probabilities:

$$\mathbb{E}(X) = \sum_x x \cdot P(X = x).$$

Example 5: A fair die

Each value $1, \ldots, 6$ has probability $1/6$, so

$$\mathbb{E}(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + \cdots + 6 \cdot \tfrac{1}{6} = 3.5.$$

With equal weights we get the ordinary average. The result shows two things:

1. 3.5 is not a value the die can show, so the expected value is not "the value we expect to see";
2. it is rather the *centre* around which the values spread: averaging a very large number of rolls gives about 3.5.

Example 6: A lottery

A ticket wins 10 € with probability 0.1 and nothing with probability 0.9:

$$\mathbb{E}(X) = 10 \cdot 0.1 + 0 \cdot 0.9 = 1\,€.$$

The plain average of the two values, $(10 + 0)/2 = 5$ €, would ignore that winning is rare. Playing many times, we win 1 € per game on average, so a ticket costing more than 1 € loses money in the long run.

Using the expected value to decide whether a deal pays off is exactly what the next example does.

Example 7: A fair price for a stock

You have 18 € and consider buying a share that costs 18 € and holding it for one year. After one year the share is worth

$$\text{Value} = 18 + X,$$

where X is the change of the price over the year. The alternative is the bank at 5% interest, which returns $18 \cdot 1.05 = 18.90$ € for sure, a certain gain of 0.90 €.

Today X is unknown: it is a random variable, and a sure number cannot be compared with a random quantity. We need a *reference value* for X, a single number representing it, which is the summary problem we started from.

The summary we choose is the gain we "expect", $\mathbb{E}(X)$: if $\mathbb{E}(X) > 0.90$ € the share pays more than the bank on average, otherwise the bank does.

To see it with numbers, suppose that the price change has distribution

x (€)	-2	0	1	3
$P(X = x)$	0.1	0.4	0.3	0.2

Then $\mathbb{E}(X) = (-2)(0.1) + 0(0.4) + 1(0.3) + 3(0.2) = 0.70$ €, less than 0.90 €: on average the bank pays more. The 20% chance of gaining 3 € is weighed together with the chance of losing and of gaining nothing.

The **fair price** p of the share is the one at which share and bank pay the same on average. Here the share is worth 18.70 € on average after one year, so

$$\underbrace{p \cdot 1.05}_{\substack{p \text{ in the bank} \\ \text{after one year}}} = \underbrace{18.70}_{\substack{\text{the share on average} \\ \text{after one year}}}$$

gives $p \approx 17.81$ €: at 18 € the share is slightly overpriced.

The expected value looks only at the average and ignores the *risk*: the bank gives 0.90 € for sure, while the share may lose 2 €. Measuring how far X strays from $\mathbb{E}(X)$ will need another concept called the variance.

2.2 The definition

The intuitive idea becomes precise once we say what "weight" means for discrete and for continuous variables.

Definition 2.2: Expected value

The **expected value** (expectation, mean, first moment) of $X \sim F_X$ is

$$\mathbb{E}(X) = \begin{cases} \displaystyle\sum_x x\, p_X(x) & \text{if } X \text{ is discrete with PMF } p_X, \\ \displaystyle\int_{\mathbb{R}} x\, f_X(x)\, \mathrm{d}x & \text{if } X \text{ is continuous with PDF } f_X, \end{cases}$$

provided the sum or the integral is well defined, that is $\sum_x |x|\, p_X(x) < \infty$ or $\int_{\mathbb{R}} |x|\, f_X(x)\, \mathrm{d}x < \infty$.

Both cases can be written at once as

$$\mathbb{E}(X) = \int_{\mathbb{R}} x\, \mathrm{d}F_X(x),$$

where $\mathrm{d}F_X(x)$ stands for $f_X(x)\, \mathrm{d}x$ in the continuous case, and the integral becomes the sum $\sum_x x\, p_X(x)$ in the discrete case.

What goes in, and what comes out? In goes the whole distribution F_X, which is a function; out comes a single number. A map that takes a function and returns a number is called a **functional**, so we can write

$$\mathbb{E}(X) = g(F_X), \qquad g \text{ a functional.}$$

The definition also calls the expected value the *first moment*. What are the other moments?

Definition 2.3: Moments

For $k = 1, 2, 3, \ldots$, the **k-th moment** of X is the expected value of its k-th power, $\mathbb{E}(X^k)$, provided it exists. The expected value $\mathbb{E}(X)$ is the first moment.

Example 8: The uniform distribution on $[0, 1]$

Here $f_X(x) = 1$ on $[0, 1]$, so

$$\mathbb{E}(X) = \int_0^1 x \cdot 1\, \mathrm{d}x = \left[\tfrac{x^2}{2}\right]_0^1 = \tfrac{1}{2},$$

the centre of the interval, as expected from a distribution that favours no value.

2.3 Property 0: distribution specific

The name "expectation *of* X" suggests that $\mathbb{E}(X)$ depends on the random variable in every detail. Recall that X is a function attaching a number to each outcome of the experiment. The formula, however, uses only the values x and their probabilities $p_X(x)$, which is the distribution of X. It never asks *which outcome* produces which value.

Proposition 2.1: Distribution specific

$\mathbb{E}(X)$ depends only on the distribution of X: two random variables with the same distribution have the same expected value.

Example 9: Heads and tails

Toss a fair coin once and let $X = 1$ if it lands heads and 0 otherwise, $Y = 1$ if it lands tails and 0 otherwise. X and Y are never equal: when one is 1 the other is 0. Their tables of values and probabilities, however, are identical:

value	0	1
probability	1/2	1/2

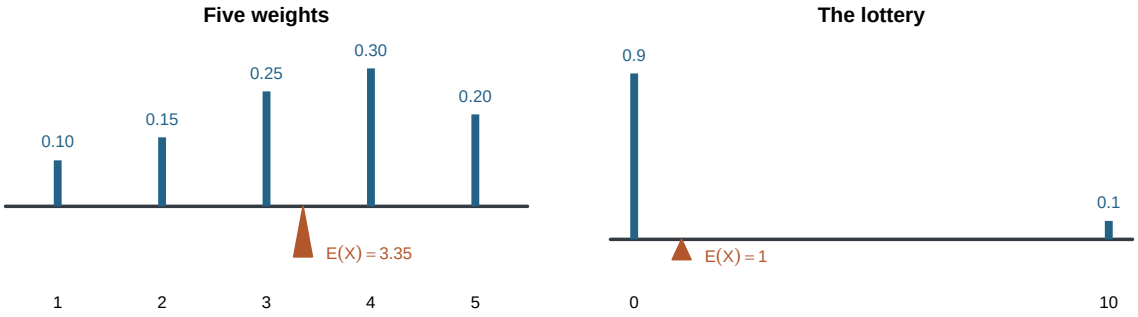
so both are Bernoulli(1/2) and $\mathbb{E}(X) = \mathbb{E}(Y) = 1/2$. The expected value cannot tell them apart, because it only sees the distribution.

2.4 Property 1: centre of mass

The expected value has a physical meaning: it is the point where the distribution balances.

2.4.1 Discrete case: a rod with weights

Picture the x-axis as a rigid, weightless rod, and hang a weight $p_X(x_j)$ at each value x_j. Resting the rod on a support placed at $\mathbb{E}(X)$, the rod stays level.



The support (triangle) at $\mathbb{E}(X)$ balances the rod.

Why exactly there? A weight on a rod makes the rod turn around the support, and its turning effect, the *moment*, grows with how heavy the weight is and how far it hangs from the support, as with a lever. With the support at c, a weight p at x has moment $(x - c)\,p$: positive if it hangs to the right and pushes the rod to the right, negative if it hangs to the left. The rod balances when the pushes cancel out:

$$\sum_j (x_j - c)\, p_X(x_j) = 0.$$

Expanding the sum and using $\sum_j p_X(x_j) = 1$ gives $\mathbb{E}(X) - c = 0$, so the support must sit at $c = \mathbb{E}(X)$.

Example 10: The lottery as a rod

Let us take the lottery of Example 6 again: $X = 0$ with probability 0.9 and $X = 10$ with probability 0.1, so $\mathbb{E}(X) = 1$. As a rod, it is a weight 0.9 hanging at 0 and a weight 0.1 hanging at 10. Does the rod really balance at 1?

With the support at $c = 1$,

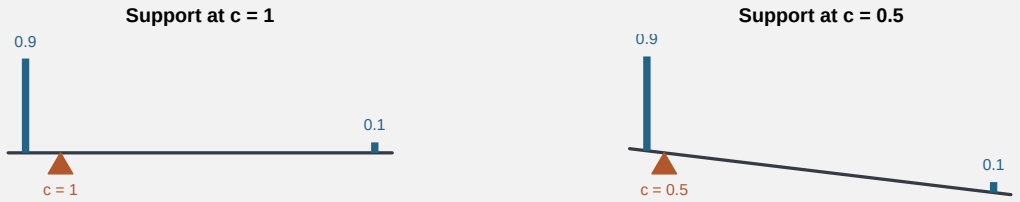
$$(0 - \underbrace{1}_{c})(0.9) + (10 - \underbrace{1}_{c})(0.1) = -0.9 + 0.9 = 0,$$

so the rod balances.

With the support at $c = 0.5$,

$$(0 - \underbrace{0.5}_{c})(0.9) + (10 - \underbrace{0.5}_{c})(0.1) = -0.45 + 0.95 = 0.5 > 0,$$

so the rod tips to the right.



The small weight is far away, and its long lever moves the balance point from 0 to 1.

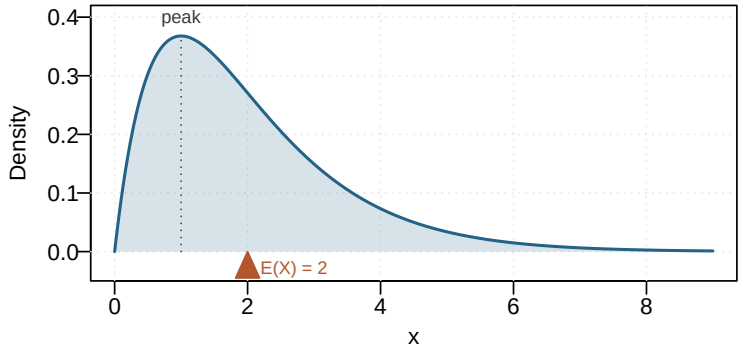
2.4.2 Continuous case: a rod with spread mass

When X is continuous no single point carries a weight; the mass is spread along the rod, like a metal bar that is thicker in some places than in others. Take the density $f_X(x)$ as the mass per unit length: where the curve is high the rod is thick and heavy, where it is low the rod is thin. The centre of gravity of this rod is $\mathbb{E}(X)$.

Cutting the rod into tiny pieces of length $\mathrm{d}x$, the piece at x has mass about $f_X(x)\,\mathrm{d}x$ and moment $(x - c)\,f_X(x)\,\mathrm{d}x$. Adding all the pieces turns the balance condition into

$$\int_{\mathbb{R}} (x - c)\,f_X(x)\,\mathrm{d}x = 0,$$

and, since $\int_{\mathbb{R}} f_X(x)\,\mathrm{d}x = 1$, gives $\mathbb{E}(X) - c = 0$, so the support must sit at $c = \mathbb{E}(X)$.



A density with a long right tail: $\mathbb{E}(X)$ lies to the right of the peak.

For a density with a long tail, $\mathbb{E}(X)$ does not lie under the peak but is pulled towards the tail: to the right of the peak for a long right tail, to the left for a long left tail. The tail carries little mass, but it is far away and pushes hard, just like the weight 0.1 at 10 in Example 10. So $\mathbb{E}(X)$ is not the most likely value, which is the peak, but the point where the distribution balances.

Proposition 2.2: Symmetric distributions

Since $\mathbb{E}(X)$ is a centre of mass, if the distribution of X is symmetric about a point x_0 *and* $\mathbb{E}(X)$ exists, then $\mathbb{E}(X) = x_0$.

The lever has a second, less pleasant consequence. A far-away weight pushes hard even when it is light, so what happens to the balance point if we move a tiny bit of probability very far from the origin?

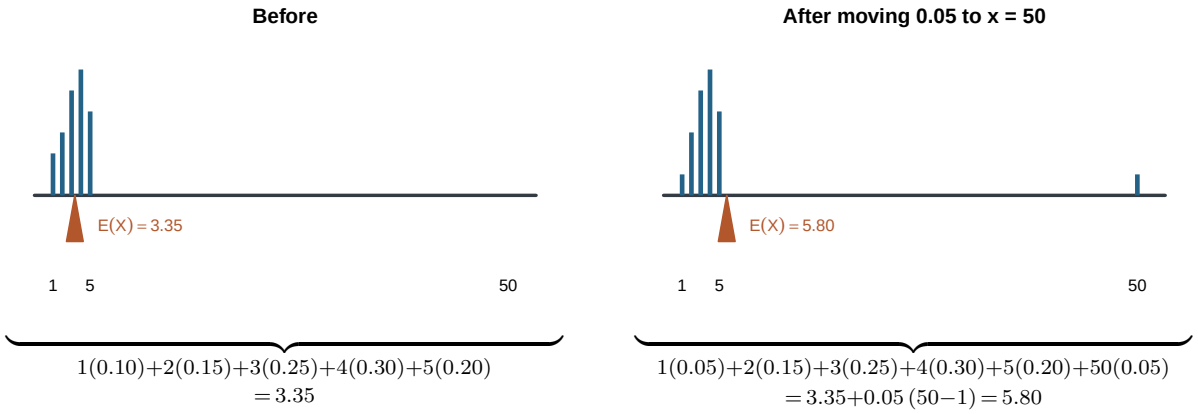
Proposition 2.3: Lack of robustness

The expected value can be affected greatly by a very small change in the probability assigned to a large value of x. Moving an amount ε of probability from a value x_j to a point M shifts $\mathbb{E}(X)$ by $\varepsilon\,(M - x_j)$: however small ε is, choosing M far enough moves $\mathbb{E}(X)$ arbitrarily far away.

Why exactly $\varepsilon\,(M - x_j)$? In $\mathbb{E}(X) = \sum_x x\,p_X(x)$ only two terms change, while the probabilities still sum to 1:

$$\begin{aligned} x_j\,p_X(x_j) &\longrightarrow x_j\,\big(p_X(x_j) - \varepsilon\big) && \text{loses } \varepsilon\,x_j, \\ M\,p_X(M) &\longrightarrow M\,\big(p_X(M) + \varepsilon\big) && \text{gains } \varepsilon\,M. \end{aligned}$$

The expected value therefore changes by $\varepsilon\,M - \varepsilon\,x_j = \varepsilon\,(M - x_j)$.



The expected value is therefore not robust to *outliers*, rare values far from the bulk of the distribution. When a model gives non-negligible probability to values very far from the centre, as the distribution of the next section does, other summaries such as the median can describe it better.

2.5 Property 2: it may not exist

The condition "and $\mathbb{E}(X)$ exists" in Proposition 2.2 is not a formality. To see why, take a distribution that looks perfectly well behaved: the Cauchy distribution of Definition 0.4.

Recall that it is symmetric around 0. So we would be tempted to say $\mathbb{E}(X) = 0$.

Let us compute it. By definition,

$$\mathbb{E}(X) = \int_{\mathbb{R}} \frac{x}{\pi\,(x^2+1)}\,\mathrm{d}x = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{2x}{x^2+1}\,\mathrm{d}x,$$

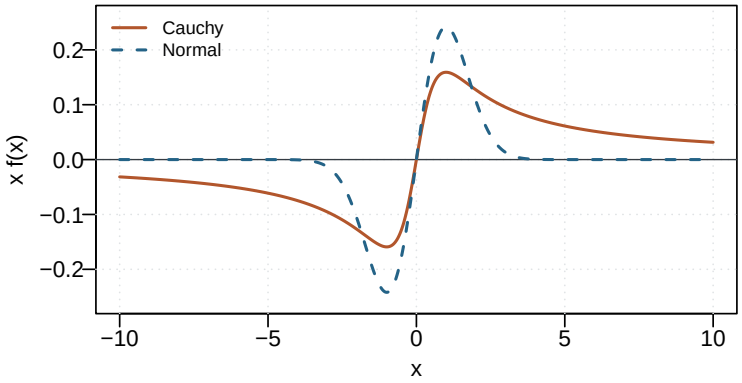
where we multiplied and divided by 2 because $2x$ is the derivative of x^2+1, so the integrand has the form $g'(x)/g(x)$ with $g(x) = x^2+1$, and a primitive is $\log(x^2+1)$.

Recall that an integral over the whole real line is defined by splitting it at a point, say 0, and computing the two halves separately; it exists only when both are finite, or at least do not diverge in opposite directions. Here

$$\int_0^{+\infty} \frac{2x}{x^2+1}\,\mathrm{d}x = \Big[\log(x^2+1)\Big]_0^{+\infty} = +\infty, \qquad \int_{-\infty}^{0} \frac{2x}{x^2+1}\,\mathrm{d}x = \Big[\log(x^2+1)\Big]_{-\infty}^{0} = -\infty,$$

and the total would be $-\infty + \infty$, which has no meaning. The expected value of a Cauchy does not exist, symmetry notwithstanding.

The reason is in the tails. The integrand $x\,f_X(x)$ behaves like $1/(\pi x)$ for large $|x|$: it goes to zero, but too slowly for its area to be finite, while for the Normal it vanishes almost immediately.



This is exactly the situation that the condition in Definition 2.2 rules out. With $|x|$ in place of x the two halves can no longer cancel: they add up, and here

$$\int_{\mathbb{R}} |x|\, f_X(x)\, \mathrm{d}x = \frac{1}{\pi} \int_0^{+\infty} \frac{2x}{x^2+1}\, \mathrm{d}x = +\infty,$$

so the condition fails and $\mathbb{E}(X)$ is not defined.

Yet the Cauchy clearly has a centre: it is symmetric around 0 and peaks there. If the expected value cannot describe it, what about the median?

2.6 Property 3: the law of the lazy statistician

Often we are not interested in X itself but in a transformation of it: its square X^2, its exponential e^X, a gain that depends on X.

Recall that we write $Y = r(X)$ for a *deterministic* function r, that is a fixed function with no randomness of its own: all the randomness of Y comes from X. How do we compute $\mathbb{E}(Y)$?

By definition, $\mathbb{E}(Y)$ is the weighted average of the values of Y, so it needs the distribution of Y. Deriving it from the distribution of X is often long and inconvenient. The following theorem says we can skip that step.

Theorem 2.1: Law of the lazy statistician

Let $X \sim F_X$ and $Y = r(X)$ for a deterministic function r. Then

$$\mathbb{E}(Y) = \mathbb{E}\big(r(X)\big) = \begin{cases} \displaystyle\sum_x r(x)\, p_X(x) & \text{if } X \text{ is discrete,} \\ \displaystyle\int_{\mathbb{R}} r(x)\, f_X(x)\, \mathrm{d}x & \text{if } X \text{ is continuous.} \end{cases}$$

Only the distribution of X is needed: each value x is transformed into $r(x)$ and weighted with the probability of x. That is why the statistician is "lazy". Why is this true? In the discrete case it is only a matter of reordering a sum.

Proof (discrete case). By definition, $\mathbb{E}(Y) = \sum_y y\, P(Y = y)$. The variable Y takes the value y whenever X takes any value x with $r(x) = y$, so

$$P(Y = y) = \sum_{x:\, r(x)=y} p_X(x).$$

Substituting, and writing y as $r(x)$ inside the inner sum, where $r(x) = y$,

$$\mathbb{E}(Y) = \sum_y \sum_{x:\, r(x)=y} r(x)\, p_X(x) = \sum_x r(x)\, p_X(x),$$

because the double sum runs over every x exactly once, grouped by the value of $r(x)$. □

The continuous case follows the same idea, but needs more tools: for a monotone differentiable r it is a change of variables in the integral, while for a general r it requires measure theory.

Example 11: The square of a die

Let X be the result of a fair die and $Y = X^2$. Without deriving the distribution of Y,

$$\mathbb{E}(X^2) = 1^2 \cdot \tfrac{1}{6} + 2^2 \cdot \tfrac{1}{6} + \cdots + 6^2 \cdot \tfrac{1}{6} = \frac{91}{6} \approx 15.17.$$

The law is everywhere, and it starts paying off at once. A first surprise: probabilities themselves are expected values. Taking as r the indicator function $\mathbb{I}_A$ of an event A gives the following result.

Corollary 2.1: Probabilities are expected values

For any event A,

$$\mathbb{E}\big(\mathbb{I}_A(X)\big) = \int_{\mathbb{R}} \mathbb{I}_A(x)\, f_X(x)\, \mathrm{d}x = \int_A f_X(x)\, \mathrm{d}x = P(X \in A).$$

As we mentioned before, the law also saves real work when the distribution of Y would be hard to find, as in the next example.

Example 12: Transformation of a Unif$(0, 1)$

Let $X \sim \mathrm{Unif}(0, 1)$, with $f_X(x) = 1$ on $(0, 1)$, and $Y = r(X) = e^X$. Finding the density of Y would require a change of variables, but the law only needs f_X:

$$\mathbb{E}(Y) \overset{\text{def}}{=} \int_{\mathbb{R}} y\, f_Y(y)\, \mathrm{d}y = \int_0^1 e^x \cdot 1\, \mathrm{d}x = \Big[e^x\Big]_0^1 = e - 1.$$

Finally, the law computes moments, which by Definition 2.3 are expected values of powers of X.

Example 13: Second moment of a Bernoulli

Let $X \sim \text{Ber}(p)$, so $X = 1$ with probability p and $X = 0$ with probability $1 - p$. With $r(x) = x^2$,

$$\mathbb{E}(X^2) = 0^2 \cdot (1-p) + 1^2 \cdot p = p.$$

Since $0^2 = 0$ and $1^2 = 1$, here $X^2 = X$, so the second moment equals the first.

2.7 Property 4: Jensen inequality

The law of the lazy statistician computes $\mathbb{E}\big(r(X)\big)$ by transforming every value x into $r(x)$ and then averaging. An even lazier shortcut is tempting: average first, obtaining $\mathbb{E}(X)$, and transform only that number, obtaining $r\big(\mathbb{E}(X)\big)$. Do the two agree?

In general they do not:

$$\mathbb{E}\big(r(X)\big) \neq r\big(\mathbb{E}(X)\big).$$

Example 14: The square of a die, again

Let us take the die of Example 11 again, with $r(x) = x^2$: there we found $\mathbb{E}(X^2) = 91/6 \approx 15.17$, while $\mathbb{E}(X) = 3.5$. Squaring the average gives

$$\big(\mathbb{E}(X)\big)^2 = 3.5^2 = 12.25 \neq 15.17 \approx \mathbb{E}(X^2).$$

There is, however, a family of functions for which the two always agree.

Proposition 2.4: Linear functions

If $r(x) = \alpha + \beta x$ for constants α and β, then

$$\mathbb{E}\big(r(X)\big) = \mathbb{E}(\alpha + \beta X) = \alpha + \beta\,\mathbb{E}(X) = r\big(\mathbb{E}(X)\big).$$

Proof (continuous case). By the law of the lazy statistician and the linearity of the integral,

$$\mathbb{E}(\alpha + \beta X) = \int_{\mathbb{R}} (\alpha + \beta x)\, f_X(x)\,\mathrm{d}x = \alpha \int_{\mathbb{R}} f_X(x)\,\mathrm{d}x + \beta \int_{\mathbb{R}} x\, f_X(x)\,\mathrm{d}x = \alpha + \beta\,\mathbb{E}(X),$$

since the density has total area 1. □

The discrete case is the same, with sums in place of integrals.

Example 15: Celsius and Fahrenheit

A temperature C in degrees Celsius becomes $F = 32 + 1.8\,C$ in degrees Fahrenheit. If $\mathbb{E}(C) = 20$, then $\mathbb{E}(F) = 32 + 1.8 \cdot 20 = 68$.

The linearity of the integral only lets constants and sums out of it, so the equality holds *only* for linear functions: with x^2, e^x or $\log x$ nothing can be taken out.

For a non-linear r the two quantities

$$\mathbb{E}\big(r(X)\big) \qquad \text{and} \qquad r\big(\mathbb{E}(X)\big)$$

are in general different. Is there still a general relationship between them?

For convex functions there is. Before stating it, let us see what convex means.

2.7.1 Convex functions

Intuitively, a smooth function is convex when its graph always bends upwards: it has a positive curvature at every point, like a bowl. For functions with a second derivative, the curvature is exactly what the second derivative measures.

Proposition 2.5: Convexity and second derivative

If r is twice differentiable, then r is convex on $S \subseteq \mathbb{R}$ if and only if $r''(x) \geq 0$ for all $x \in S$.

For example, x^2 is convex on $\mathbb{R}$, since its second derivative is $2 > 0$, and so is e^x, whose second derivative is $e^x > 0$.

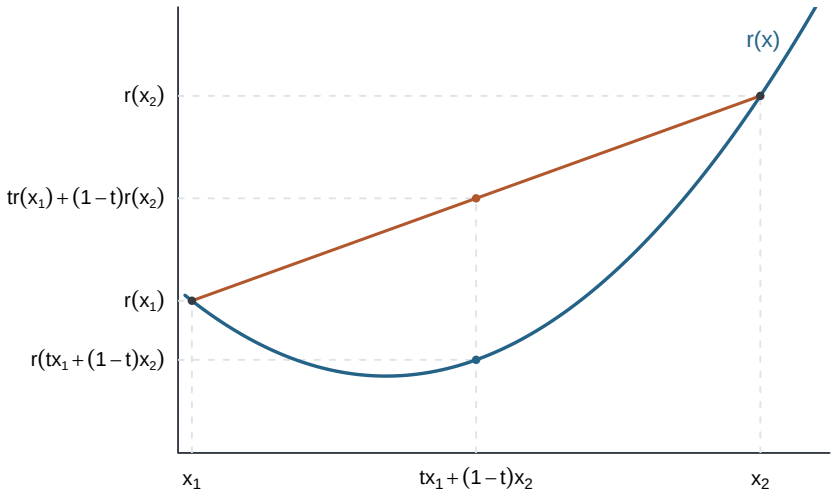
The same idea has a geometric form that needs no derivatives: if we join two points of the graph with a straight segment, the segment lies on or above the graph. The points between x_1 and x_2 are written as $t\,x_1 + (1-t)\,x_2$ with $t \in [0,1]$, and at such a point the segment has height $t\,r(x_1) + (1-t)\,r(x_2)$.

Definition 2.4: Convex function

A function $r : \mathbb{R} \to \mathbb{R}$ is **convex** on $S \subseteq \mathbb{R}$ if, for any $x_1, x_2 \in S$ and $t \in [0,1]$,

$$r\big(t\,x_1 + (1-t)\,x_2\big) \leq t\,r(x_1) + (1-t)\,r(x_2).$$

It is **concave** if $-r$ is convex.



For a concave function the segment lies below the graph and $r''(x) \leq 0$: for example $\log x$, whose second derivative is $-1/x^2 < 0$.

2.7.2 Jensen's inequality

We can now answer the question for convex functions.

Theorem 2.2: Jensen's inequality

Let $X \sim F_X$ and let r be a convex function, with $\mathbb{E}(X)$ and $\mathbb{E}\big(r(X)\big)$ existing. Then

$$r\big(\mathbb{E}(X)\big) \leq \mathbb{E}\big(r(X)\big).$$

If r is concave, the inequality is reversed.

Jensen's inequality is the definition of convexity rewritten in probabilistic terms, as the case of a variable with two values shows.

Proof (two-valued case). Take x_1, x_2 and $t \in [0,1]$, and define a discrete random variable X that takes two values,

$$P(X = x_1) = t, \qquad P(X = x_2) = 1 - t.$$

Its expected value is $\mathbb{E}(X) = t\,x_1 + (1-t)\,x_2$, and by the law of the lazy statistician

$$\mathbb{E}\big(r(X)\big) = t\,r(x_1) + (1-t)\,r(x_2).$$

Since r is convex, Definition 2.4 gives

$$\underbrace{r\big(t\,x_1 + (1-t)\,x_2\big)}_{r(\mathbb{E}(X))} \;\leq\; \underbrace{t\,r(x_1) + (1-t)\,r(x_2)}_{\mathbb{E}(r(X))},$$

which is exactly $r\big(\mathbb{E}(X)\big) \leq \mathbb{E}\big(r(X)\big)$. □

We will see the proof for a general X later.

Let us take the die of Example 11 again: there $r(x) = x^2$ is convex, and indeed $\big(\mathbb{E}(X)\big)^2 = 12.25 \leq 15.17 \approx \mathbb{E}(X^2)$.

Example 16: Arithmetic and geometric mean

Since $\log x$ is concave, Jensen's inequality gives $\log \mathbb{E}(X) \geq \mathbb{E}(\log X)$ for any positive random variable X. Take data $x_1, \ldots, x_n > 0$ and their empirical distribution, the random variable X with $P(X = x_i) = 1/n$ for each i. Then

$$\mathbb{E}(X) = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}_n, \qquad \mathbb{E}(\log X) = \frac{1}{n}\sum_{i=1}^{n} \log x_i = \log\Big(\prod_{i=1}^{n} x_i\Big)^{1/n},$$

so Jensen's inequality reads

$$\log\Big(\frac{1}{n}\sum_{i=1}^{n} x_i\Big) \geq \log\Big(\prod_{i=1}^{n} x_i\Big)^{1/n} \quad\Longleftrightarrow\quad \frac{1}{n}\sum_{i=1}^{n} x_i \geq \sqrt[n]{\prod_{i=1}^{n} x_i},$$

where we exponentiated both sides, which keeps the inequality because e^x is increasing. The arithmetic mean is always at least the geometric mean: for 1 and 4, $2.5 \geq \sqrt{4} = 2$.

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Jensen's inequality also gives back Proposition 2.4. What happens if r is both convex and concave? Then the segment between two points of the graph lies both above and below the graph, so it lies exactly on it:

$$r\big(t\,x_1 + (1-t)\,x_2\big) = t\,r(x_1) + (1-t)\,r(x_2).$$

Every chord coincides with the graph, so the graph is a straight line. With derivatives, $r'' \geq 0$ and $r'' \leq 0$ give $r'' = 0$, that is $r(x) = \alpha + \beta x$. Jensen's inequality now holds in both directions: since r is convex, $r\big(\mathbb{E}(X)\big) \leq \mathbb{E}\big(r(X)\big)$, and since r is concave, $r\big(\mathbb{E}(X)\big) \geq \mathbb{E}\big(r(X)\big)$. Together they give

$$\mathbb{E}(\alpha + \beta X) = \alpha + \beta\,\mathbb{E}(X),$$

which is Proposition 2.4 again.

The last example compares the expected value with the median. For well-behaved random variables the two cannot be too far apart, and Jensen's inequality tells us how far.

Example 17: Expected value and median

Let $\mu = \mathbb{E}(X)$, assuming it exists, and let m be the median of X. With the convex function $r(x) = |x|$,

$$\begin{aligned}
|\mu - m| &= \big|\mathbb{E}(X) - m\big| = \big|\mathbb{E}(X - m)\big| && \text{linear functions}\\
&\leq \mathbb{E}\,|X - m| && \text{Jensen, } |x| \text{ convex}\\
&\leq \mathbb{E}\,|X - \mu| && (\heartsuit)\\
&= \mathbb{E}\sqrt{(X-\mu)^2}\\
&\leq \sqrt{\mathbb{E}(X-\mu)^2} && \text{Jensen, } \sqrt{x} \text{ concave.}
\end{aligned}$$

Step ($\heartsuit$) uses a property of the median that we will prove in Property 5: among all constants c, the median minimises $\mathbb{E}\,|X - c|$. The last quantity, $\sigma = \sqrt{\mathbb{E}(X-\mu)^2}$, is the **standard deviation**, which we will meet again with the variance. Hence

$$|\mu - m| \leq \sigma :$$

the expected value and the median are at most one standard deviation apart.

2.7.3 Jensen and the ELBO

So far we have used Jensen's inequality to compare summaries of a distribution. It also has an important use in learning, where it gives the **evidence lower bound**, or **ELBO**. To see it, we need to talk a little about learning again.

Recall from Section 1.2 that we model what we observe together with a hidden variable, through a joint distribution $p(\text{obs}, \text{hid}) = p(\text{obs} \mid \text{hid})\, p(\text{hid})$, and that the distribution of what we see is the mixture

$$p(\text{obs}) = \int_{\text{supp(hid)}} p(\text{obs}, \text{hid})\, \mathrm{d}\,\text{hid}.$$

Learning θ from the likelihood of such a mixture is hard: for a GMM without labels there is no formula for $\hat{\theta}$, which is problem (♠). It would be easy if we knew the group of every observation, since we could use Proposition 1.2. But now we have learned Jensen's inequality: could it help us solve problem (♠)?

The trouble is that we want to maximise the logarithm of an integral:

$$\log p_\theta(\text{obs}) = \log \int_{\text{supp(hid)}} p_\theta(\text{obs}, \text{hid})\, \mathrm{d}\,\text{hid}.$$

The idea is to push back the hidden variable. Take any distribution q over the hidden variable, and multiply and divide by it inside the integral, which changes nothing since $q/q = 1$:

$$\log p_\theta(\text{obs}) = \log \int_{\text{supp(hid)}} \underbrace{q(\text{hid})}_{\text{density}} \underbrace{\frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})}}_{r(\text{hid})}\, \mathrm{d}\,\text{hid}.$$

By the law of the lazy statistician:

$$\log p_\theta(\text{obs}) = \log \mathbb{E}_q\big(r(\text{hid})\big) = \log \mathbb{E}_q\left[\frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})}\right].$$

We now have the logarithm of an expected value, and log is concave: this is exactly the situation of Jensen's inequality.

By Jensen's inequality:

$$\log p_\theta(\text{obs}) = \log \mathbb{E}_q\left[\frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})}\right] \;\geq\; \mathbb{E}_q\left[\log \frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})}\right] =: F(q, \theta).$$

What have we gained? On the left we still have the logarithm of a sum, the one that blocked us in problem (♠). On the right the logarithm has moved inside the expected value, so we take the logarithm of each term instead of the logarithm of their sum.

For a GMM the terms are $\log\big(\pi_j\, N(o \mid \mu_j, \sigma_j^2)\big)$, which split into simple pieces, as in the case with labels of Proposition 1.2.

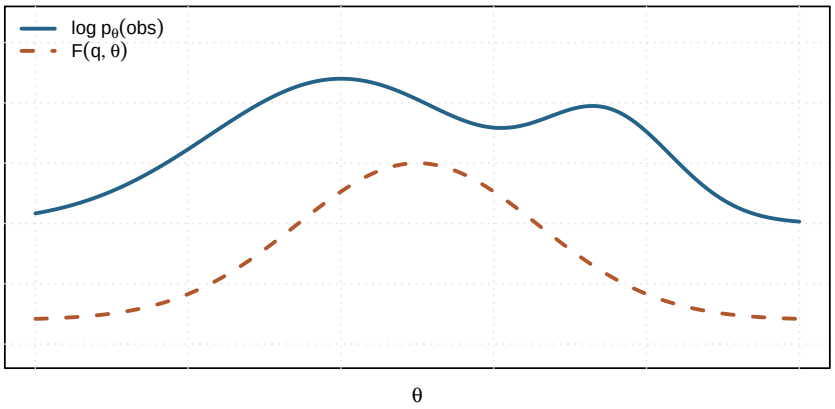
Definition 2.5: ELBO

Let obs be the observed variable, hid the hidden one, and q any distribution of hid. The **evidence lower bound** (ELBO) is

$$F(q, \theta) = \mathbb{E}_q \left[\log \frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})} \right].$$

Evidence refers to $\log p_\theta(\text{obs})$, the log-likelihood of what we observe, and *lower bound* to the fact that $F(q, \theta) \leq \log p_\theta(\text{obs})$ for every q and every θ.

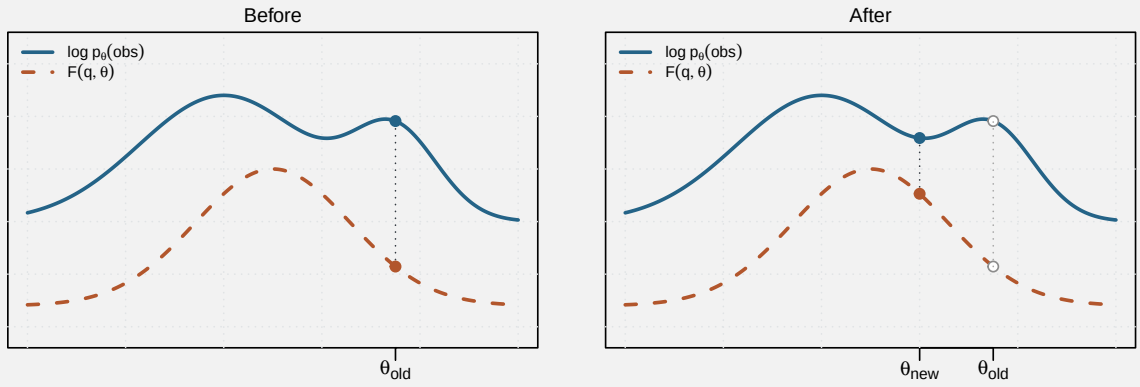
The ELBO is our **surrogate target**: instead of maximising $\log p_\theta(\text{obs})$, which is hard, as problem (♠) shows for a GMM, we maximise $F(q, \theta)$, which is easier. In the plot below the curve $F(q, \theta)$ never rises above $\log p_\theta(\text{obs})$, so by pushing F up we push up a lower bound of the curve we care about.



But does raising F really raise $\log p_\theta(\text{obs})$ as well? Not necessarily, as the next example shows.

Example 18: Raising F while $\log p_\theta(\text{obs})$ goes down

Keep q fixed and take the two curves of the plot above. We start from θ_{old}, where $\log p_\theta(\text{obs})$ is close to its highest point and $F(q, \theta)$ lies far below it. Then we move to θ_{new}, chosen so that F goes up. In the plot on the right the hollow points mark where we were: the orange point has risen, but the blue one has gone down. Raising F has lowered the log-likelihood.



Whether raising F raises $\log p_\theta(\text{obs})$ depends on how wide the gap between the two curves is: at θ_{old} the gap was so wide that F could rise a lot without telling us anything about $\log p_\theta(\text{obs})$.

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2.7.4 The information gap

How wide is the gap between the two curves? Let us give it a name and compute it.

Definition 2.6: Information gap

The **information gap** is the difference between the log-likelihood and the ELBO,

$$\log p_\theta(\text{obs}) - F(q, \theta).$$

By Jensen's inequality it is never negative. To see what it is, we write F as an integral and split the joint density with the multiplication rule, $p_\theta(\text{obs}, \text{hid}) = p_\theta(\text{hid} \mid \text{obs})\, p_\theta(\text{obs})$:

$$\begin{aligned} \log p_\theta(\text{obs}) - F(q, \theta) &= \log p_\theta(\text{obs}) - \int q(\text{hid}) \, \log \frac{p_\theta(\text{obs}, \text{hid})}{q(\text{hid})} \, \mathrm{d}\,\text{hid} \\ &= \log p_\theta(\text{obs}) - \int q(\text{hid}) \, \log \frac{p_\theta(\text{hid} \mid \text{obs})\, p_\theta(\text{obs})}{q(\text{hid})} \, \mathrm{d}\,\text{hid}. \end{aligned}$$

The logarithm of a product is the sum of the logarithms, so the integral splits in two:

$$= \log p_\theta(\text{obs}) - \left[\int q(\text{hid}) \, \log \frac{p_\theta(\text{hid} \mid \text{obs})}{q(\text{hid})} \, \mathrm{d}\,\text{hid} + \int q(\text{hid}) \, \log p_\theta(\text{obs}) \, \mathrm{d}\,\text{hid} \right].$$

In the second integral $\log p_\theta(\text{obs})$ does not depend on hid, so it comes out, and what is left integrates to 1 because q is a distribution:

$$\int q(\text{hid}) \, \log p_\theta(\text{obs}) \, \mathrm{d}\,\text{hid} = \log p_\theta(\text{obs}) \underbrace{\int q(\text{hid}) \, \mathrm{d}\,\text{hid}}_{1} = \log p_\theta(\text{obs}).$$

The two copies of $\log p_\theta(\text{obs})$ cancel, and turning the fraction upside down changes the sign of the logarithm:

$$\log p_\theta(\text{obs}) - F(q, \theta) = \int q(\text{hid}) \, \log \frac{q(\text{hid})}{p_\theta(\text{hid} \mid \text{obs})} \, \mathrm{d}\,\text{hid}.$$

This integral is a very specific way of measuring how far q is from the distribution of the hidden variable given what we observe.

Definition 2.7: Kullback–Leibler divergence

The **Kullback–Leibler divergence** (KL) of q from p is

$$\mathrm{KL}(q \mid p) = \int q(x) \, \log \frac{q(x)}{p(x)} \, \mathrm{d}x.$$

It is not a distance, and $\mathrm{KL}(q \mid p) = 0$ if and only if $q = p$.

So the information gap is

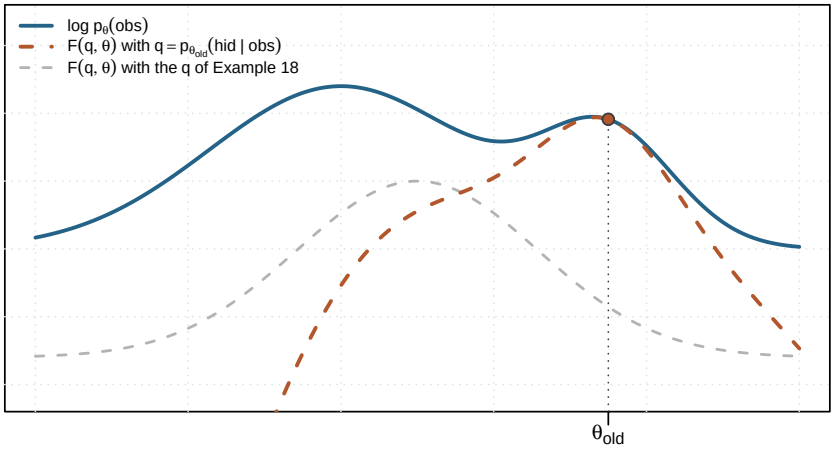
$$\log p_\theta(\text{obs}) - F(q, \theta) = \mathrm{KL}\big(q(\text{hid}) \mid p_\theta(\text{hid} \mid \text{obs})\big),$$

and it is zero exactly when

$$q(\text{hid}) = p_\theta(\text{hid} \mid \text{obs}).$$

This answers our question: if we choose q as the distribution of the hidden variable given what we observe, the two curves touch, and $F(q, \theta) = \log p_\theta(\text{obs})$ at that θ.

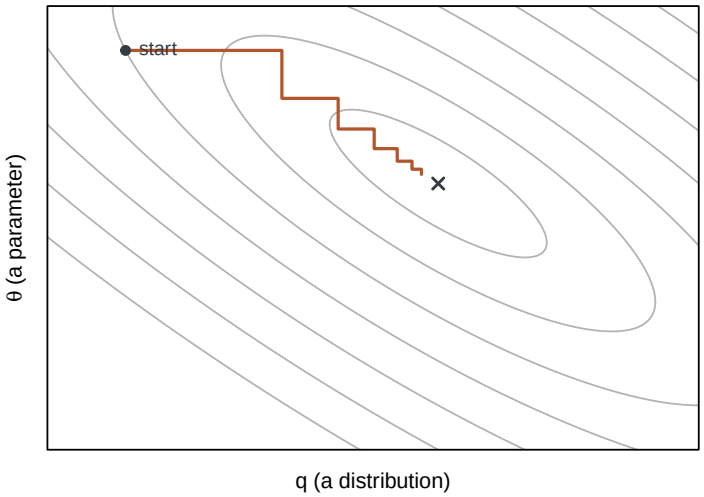
In the plot below we take θ_{old} of Example 18 again and choose $q = p_{\theta_{\text{old}}}(\text{hid} \mid \text{obs})$.



How can we use this to climb $\log p_\theta(\text{obs})$?

2.7.5 The EM algorithm

The ELBO $F(q, \theta)$ depends on two things we can move: the distribution q and the parameter θ. The typical way to use it is the **EM algorithm**, one example of the wider family of MM algorithms. The idea is **coordinate ascent** over $F(q, \theta)$: we move one coordinate at a time, keeping the other fixed. First we optimise over q with θ fixed, then over θ with q fixed, and we repeat.



In the plot each grey curve joins the points with the same value of F, and the cross marks its maximum. A horizontal step changes only q, a vertical step changes only θ, and each step climbs as high as it can.

Algorithm 2.1: EM algorithm

Start from a parameter $\theta^{(0)}$. For $t = 1, 2, \ldots$ repeat:

1. **E-step.** Optimise over all distributions q, with θ fixed:
$$q^{(t)} = \arg\max_{q} F\big(q, \theta^{(t-1)}\big).$$
2. **M-step.** Optimise over all parameters θ, with q fixed:
$$\theta^{(t)} = \arg\max_{\theta} F\big(q^{(t)}, \theta\big).$$

E-step

Which q does the E-step choose? Recall from Section 2.7.4 that

$$F(q, \theta) = \log p_\theta(\text{obs}) - \mathrm{KL}\big(q(\text{hid}) \mid p_\theta(\text{hid} \mid \text{obs})\big).$$

With $\theta = \theta^{(t-1)}$ fixed, the first term does not depend on q, so F is largest when the KL is 0, that is, when

$$q^{(t)}(\text{hid}) = p_{\theta^{(t-1)}}(\text{hid} \mid \text{obs}),$$

if we can compute it.

M-step

The M-step, instead, depends on the model: for a GMM we will see it in the next lecture.

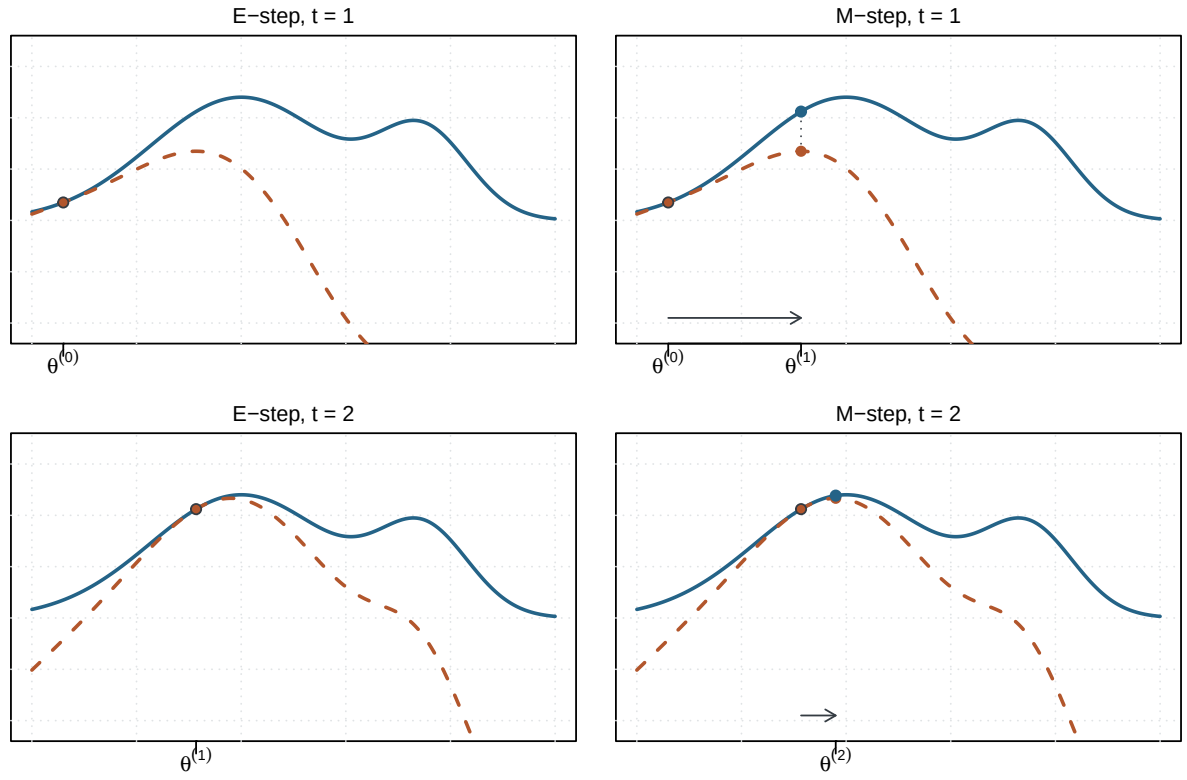
Whatever form the M-step takes, does one round of EM really climb $\log p_\theta(\text{obs})$, the target we want to maximise? Let us follow one round. At the beginning we are at $\log p_{\theta^{(t-1)}}(\text{obs})$. Then:

$$\log p_{\theta^{(t-1)}}(\text{obs}) \underbrace{=}_{\text{E-step}} F\big(q^{(t)}, \theta^{(t-1)}\big) \underbrace{\leq}_{\text{M-step}} F\big(q^{(t)}, \theta^{(t)}\big) \underbrace{\leq}_{\text{Jensen}} \log p_{\theta^{(t)}}(\text{obs}).$$

- The E-step closes the gap, so F equals the log-likelihood at $\theta^{(t-1)}$: the two curves touch, as in the last plot.
- The M-step chooses the θ with the largest F, so F can only go up.
- By Jensen's inequality F never rises above the log-likelihood.

So every round of EM increases, or at least does not decrease, the original objective $\log p_\theta(\text{obs})$.

What does this look like round after round? In the sketch below we start from $\theta^{(0)}$ and run two rounds of EM. In each E-step a new orange curve touches the blue one at the current θ; in each M-step we move to the top of the orange curve, and the blue curve is higher there.



The second step is shorter than the first: as EM approaches a maximum of $\log p_\theta(\text{obs})$, each round gains less.

Remarks. Both steps of Algorithm 2.1 ask for a maximum. Do we always get it, and do we really need it?

1. **E-step.** Optimising over *all* distributions q is sometimes not feasible: sometimes $p_\theta(\text{hid} \mid \text{obs})$ is not available. What do we do then? We optimise over a smaller class of distributions instead of the whole class. This is the idea of **variational methods**.
2. **M-step.** Here we are fine even without the exact maximum. In the chain above the M-step is only used through $F\big(q^{(t)}, \theta^{(t-1)}\big) \le F\big(q^{(t)}, \theta^{(t)}\big)$, so any $\theta^{(t)}$ that does not lower F still makes $\log p_\theta(\text{obs})$ climb.

What does the M-step maximise? Let us split the ELBO in two. The logarithm of a ratio is the difference of the logarithms, so

$$F(q,\theta) = \mathbb{E}_q\big[\log p_\theta(\text{obs},\text{hid})\big] - \mathbb{E}_q\big[\log q(\text{hid})\big] = \underbrace{\mathbb{E}_q\big[\log p_\theta(\text{obs},\text{hid})\big]}_{\text{expected complete log-likelihood}} + \underbrace{H(q)}_{\text{entropy of } q},$$

where $H(q) = -\mathbb{E}_q\big[\log q(\text{hid})\big]$ is called the **entropy** of q. In the first term, $\log p_\theta(\text{obs},\text{hid})$ is the **complete log-likelihood**, the one we would have if we also observed the hidden variable, and we average it over q. The entropy depends only on q, and in the M-step $q = q^{(t)}$ is fixed, so

$$\theta^{(t)} = \arg\max_\theta F\big(q^{(t)},\theta\big) = \arg\max_\theta \mathbb{E}_{q^{(t)}}\big[\log p_\theta(\text{obs},\text{hid})\big].$$

The M-step maximises the expected complete log-likelihood: it works as if we knew the hidden variable, weighting each of its values by $q^{(t)}$.

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2.7.6 EM for a Gaussian mixture

So far we have presented the EM algorithm without going into the details, in particular of the M-step.

Let us now run it on a concrete model: the GMM without labels, the one of problem (♠). We have the data $o_1, \dots, o_n$, a number of groups K that we keep fixed, and the likelihood

$$L(\theta) = \prod_{i=1}^{n} \underbrace{\sum_{j=1}^{K} \pi_j \, N(o_i \mid \mu_j, \sigma_j^2)}_{p_\theta(o_i)}, \qquad \theta = \{\pi_j, \mu_j, \sigma_j^2\}_{j=1}^K.$$

What do the E-step and the M-step of Algorithm 2.1 become here?

E-step

As we saw in the EM algorithm, the optimal choice is $q(\text{hid}) = p_\theta(\text{hid} \mid \text{obs})$, if we can compute it. Let us see if we can. Each observation o_i has its own hidden variable hid_i, the group it comes from, and by Bayes' theorem its posterior is

$$p_\theta(\text{hid}_i = j \mid o_i) = \frac{p_\theta(o_i \mid \text{hid}_i = j)\, p_\theta(\text{hid}_i = j)}{p_\theta(o_i)} = \frac{\pi_j\, N(o_i \mid \mu_j, \sigma_j^2)}{\sum_{k=1}^{K} \pi_k\, N(o_i \mid \mu_k, \sigma_k^2)}.$$

Once θ is fixed every piece is a number we can compute, and the denominator is a sum of only K terms, so for a GMM we can.

Notice that, since the denominator does not depend on j,

$$p_\theta(\text{hid}_i = j \mid o_i) \propto p_\theta(o_i \mid \text{hid}_i = j)\, p_\theta(\text{hid}_i = j).$$

Definition 2.8: Responsibilities

The **responsibility** of group j for the observation o_i is the posterior probability of its hidden variable,

$$r_j^{(i)} = p_\theta(\text{hid}_i = j \mid o_i), \qquad i = 1, \dots, n, \quad j = 1, \dots, K.$$

Example 4 part 6: Heights of men and women

Let us take the heights of Example 4 part 5 again. With the labels we found $\hat\theta_{\text{women}} \approx (160.5,\ 7.1^2)$, $\hat\theta_{\text{men}} \approx (176.3,\ 7.4^2)$ and weights $\hat\pi_1 = \hat\pi_2 = 0.5$. Suppose these are the current parameters of EM. What responsibilities do the heights 147, 167 and 184 get?

Using Bayes' theorem as above:

height o_i	$r^{(i)}_{\text{women}}$	$r^{(i)}_{\text{men}}$
147	0.998	0.002
167	0.60	0.40
184	0.007	0.993

A very short or very tall person is almost surely in one group. A height of 167 cm, between the two centres, is shared: 60% women and 40% men. Every observation gets a weight for every group, instead of a single label.

M-step

Once we have the responsibilities, the M-step looks for the new parameters θ. Recall that it maximises the expected complete log-likelihood:

$$\theta^{(t)} = \arg\max_\theta \mathbb{E}_{q^{(t)}}\big[\log p_\theta(\text{obs}, \text{hid})\big].$$

Let us call the function to maximise $E(\theta) = \mathbb{E}_{q^{(t)}}\big[\log p_\theta(\text{obs}, \text{hid})\big]$. For a GMM it is the double sum

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \log\big(\pi_j\, N(o_i \mid \mu_j, \sigma_j^2)\big).$$

The logarithm of a product is the sum of the logarithms, so

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \Big[\log \pi_j + \log N(o_i \mid \mu_j, \sigma_j^2)\Big].$$

Now we write the Normal density explicitly:

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \Big[\log \pi_j + \log\Big(\frac{1}{\sqrt{2\pi}\,\sigma_j}\,\exp\Big(-\frac{(o_i-\mu_j)^2}{2\sigma_j^2}\Big)\Big)\Big].$$

The second logarithm is again the logarithm of a product, so it is the sum of the logarithms of its three factors:

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \Big[\log \pi_j + \log\frac{1}{\sqrt{2\pi}} + \log\frac{1}{\sigma_j} + \log\exp\Big(-\frac{(o_i-\mu_j)^2}{2\sigma_j^2}\Big)\Big].$$

Since $\log\frac{1}{a} = -\log a$, $\log\sqrt{a} = \frac{1}{2}\log a$ and the logarithm cancels the exponential,

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \Big[\log \pi_j - \tfrac{1}{2}\log(2\pi) - \log\sigma_j - \frac{(o_i-\mu_j)^2}{2\sigma_j^2}\Big].$$

The term $-\frac{1}{2}\log(2\pi)$ does not depend on θ, so it does not change where the maximum is, and we can drop it:

$$E(\theta) = \sum_{i=1}^{n}\sum_{j=1}^{K} r_j^{(i)} \Big[\log \pi_j - \log\sigma_j - \frac{(o_i-\mu_j)^2}{2\sigma_j^2}\Big] + \text{const.}$$

At the start we could not maximise the likelihood with a closed form, because of the logarithm of a sum in problem (♠). Now the logarithm is inside the sums, and each term is as simple as in the case with labels. So now we can.

The centres. Only the terms of group j contain μ_j. Setting the derivative to zero,

$$\frac{\partial E}{\partial \mu_j} = \sum_{i=1}^{n} r_j^{(i)}\,\frac{o_i - \mu_j}{\sigma_j^2} = 0 \quad\Longleftrightarrow\quad \hat{\mu}_j = \frac{\sum_{i=1}^{n} r_j^{(i)}\, o_i}{\sum_{i=1}^{n} r_j^{(i)}}.$$

The new centre of group j is a weighted average of the observations: each one counts as much as its responsibility for group j.

The spreads. In the same way, setting to zero the derivative with respect to σ_j,

$$\frac{\partial E}{\partial \sigma_j} = \sum_{i=1}^{n} r_j^{(i)} \Big[-\frac{1}{\sigma_j} + \frac{(o_i - \mu_j)^2}{\sigma_j^3} \Big] = 0 \quad \Longleftrightarrow \quad \hat{\sigma}_j^2 = \frac{\sum_{i=1}^{n} r_j^{(i)} (o_i - \hat{\mu}_j)^2}{\sum_{i=1}^{n} r_j^{(i)}}.$$

The new $\hat{\sigma}_j^2$ is a weighted average of the squared distances from the new centre.

Compare these with maximum likelihood for a GMM with labels. If each responsibility were 1 for the true group and 0 for the others, $\sum_i r_j^{(i)}$ would be n_j, and $\hat{\mu}_j$ and $\hat{\sigma}_j^2$ would be the centre and the spread computed from the observations of group j only. The M-step is maximum likelihood with labels, where the labels 0 or 1 are replaced by the responsibilities.

Example 4 part 7: Heights of men and women

Let us take the three heights of Example 4 part 6 again: 147, 167 and 184, with responsibilities 0.998, 0.60 and 0.007 for women and 0.002, 0.40 and 0.993 for men. If these three were all our data, where would the M-step move the two centres?

For women,

$$\hat{\mu}_{\text{women}} = \frac{0.998 \cdot 147 + 0.60 \cdot 167 + 0.007 \cdot 184}{0.998 + 0.60 + 0.007} \approx \frac{248.2}{1.605} \approx 154.6,$$

and in the same way $\hat{\mu}_{\text{men}} \approx 179.1$. The height 147 pulls the women's centre almost with its full weight, 167 pulls it with weight 0.60, and 184 hardly at all.

The weights. The weights are different: they must be non-negative and add up to 1, so they lie in the simplex. Maximising $E(\theta)$ over the weights is therefore an optimisation with a constraint, which is solved with Lagrange multipliers, a tool we will see later. The result is

$$\hat{\pi}_j = \frac{\sum_{i=1}^{n} r_j^{(i)}}{n},$$

the share of the observations that group j is responsible for. With labels 0 or 1 it becomes n_j/n, as in the case with labels.

Putting it together. One round of EM for a GMM is:

1. **E-step:** with the current parameters, compute the responsibilities $r_j^{(i)}$ of every group for every observation;
2. **M-step:** with these responsibilities, compute the new centres $\hat{\mu}_j$, spreads $\hat{\sigma}_j^2$ and weights $\hat{\pi}_j$.

We repeat the two steps, and by the check of Section 2.7.5 the log-likelihood never goes down. This is the tool that problem (♠) was waiting for: without labels there is still no single formula for $\hat{\theta}$, but we can now fit a GMM by climbing its likelihood one round at a time.